Reprium

Science-Based Fortune Teller

April 28, 2025

The Matching Process: Past & Present

- Historically
  - Humans lived in *culturally homogeneous* communities
  - They evolved to optimally select partners within a familiar cultural setting
    - Based on physical, cognitive & material characteristics

The Matching Process: Past & Present

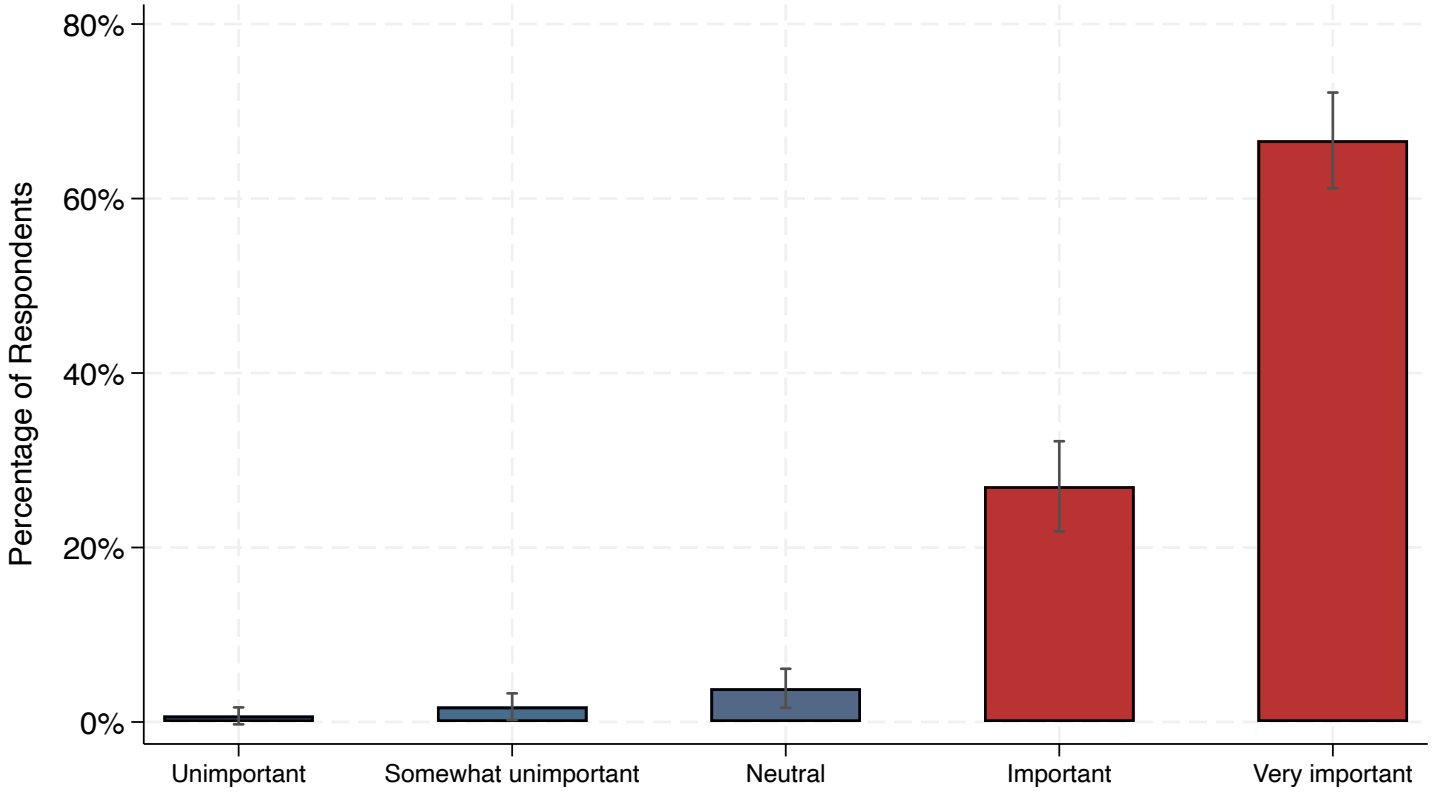
- The Modern Era
  - Rapid socioeconomic transformation
    - $\rightarrow$ Emergence of multicultural societies
  - Partner selection critically depends on the detection of cultural *synergy*
    - Yet, evolution has not equipped us to identify optimal synergy
  - We are in great need of assistance
    - We are in great need of Reprium

Cultural Synergy

- Fosters enduring and meaningful relationships
  - Bonds that endure *beyond days or weeks*
  - Lasting emotional connection
  - Potential for lifelong partnership
- Contributes to the well-being of prospective children
  - Economic prosperity
  - Academic achievement
  - Creativity
  - Focus

Lasting Relationships

Importance of Lasting Relationships



Survey Question: "When choosing a partner, how important is it for you that your relationship will be long-lasting?"

Survey conducted for Reprium by Prolific on a representative sample of the US population. Prolific is a platform that connects researchers with a diverse, global pool of participants, ensuring high-quality, reliable data through its rigorous participant vetting process and commitment to research integrity

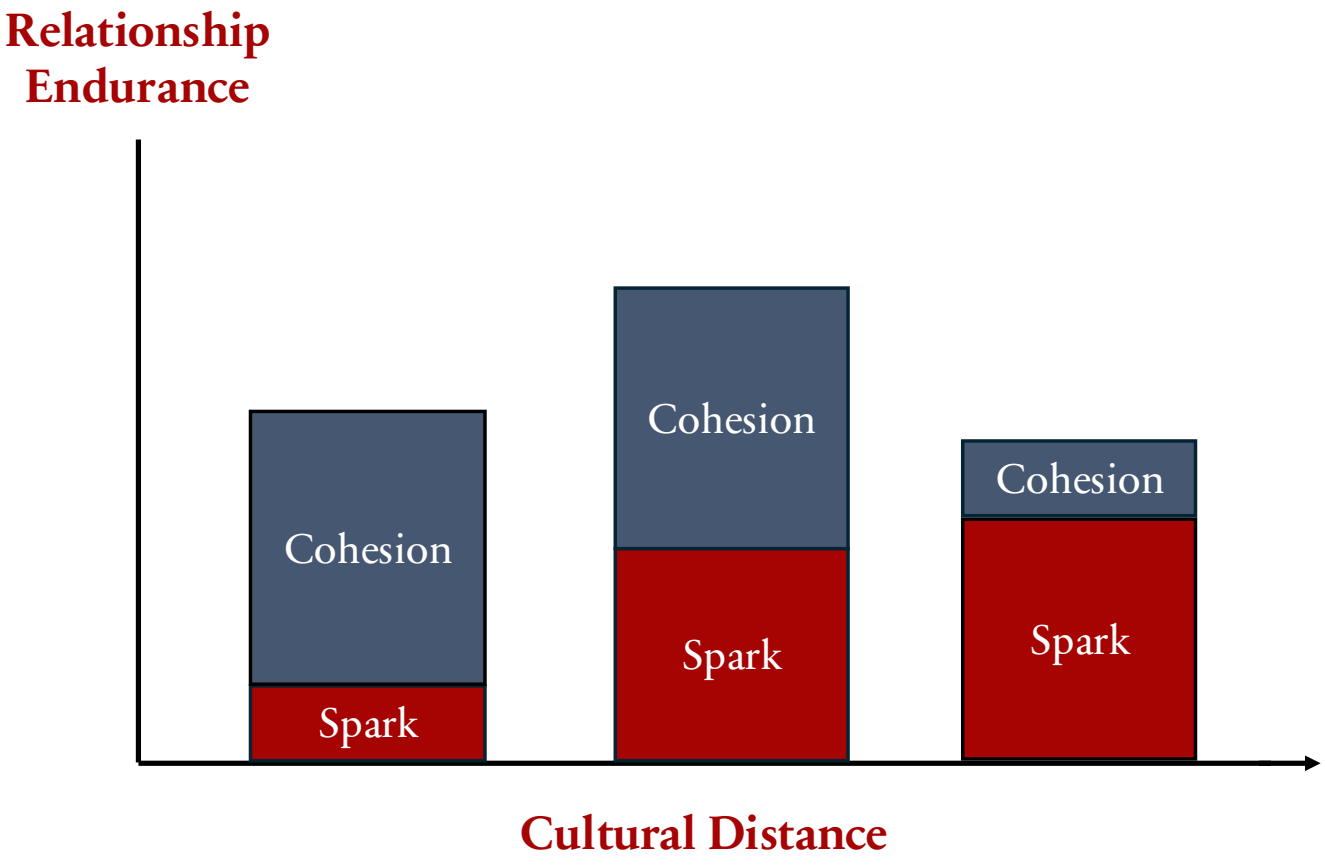
Unfulfilled Demand

- Existing dating platforms
  - Overlook the core desire for lasting relationships
    - Focus instead on match creation
  - Lack the technology to predict which matches will endure

Cultural Synergy: The Key to Lasting Relationships

- Cultural similarities tend to promote
  - Stronger cohesion (effective communication & shared values)
  - Reduced spark (familiarity & predictability)
- Cultural dissimilarities often lead to
  - Weaker cohesion (communication barriers & misaligned values)
  - Stronger spark (exploration & discovery)
- An intermediate level of cultural dissimilarity may strike the ideal balance
  - Sufficient similarities for cohesion & enough differences for spark
- This optimal blend fosters fulfilling and durable relationships
  - The foundation for lasting love

Conceptual Framework



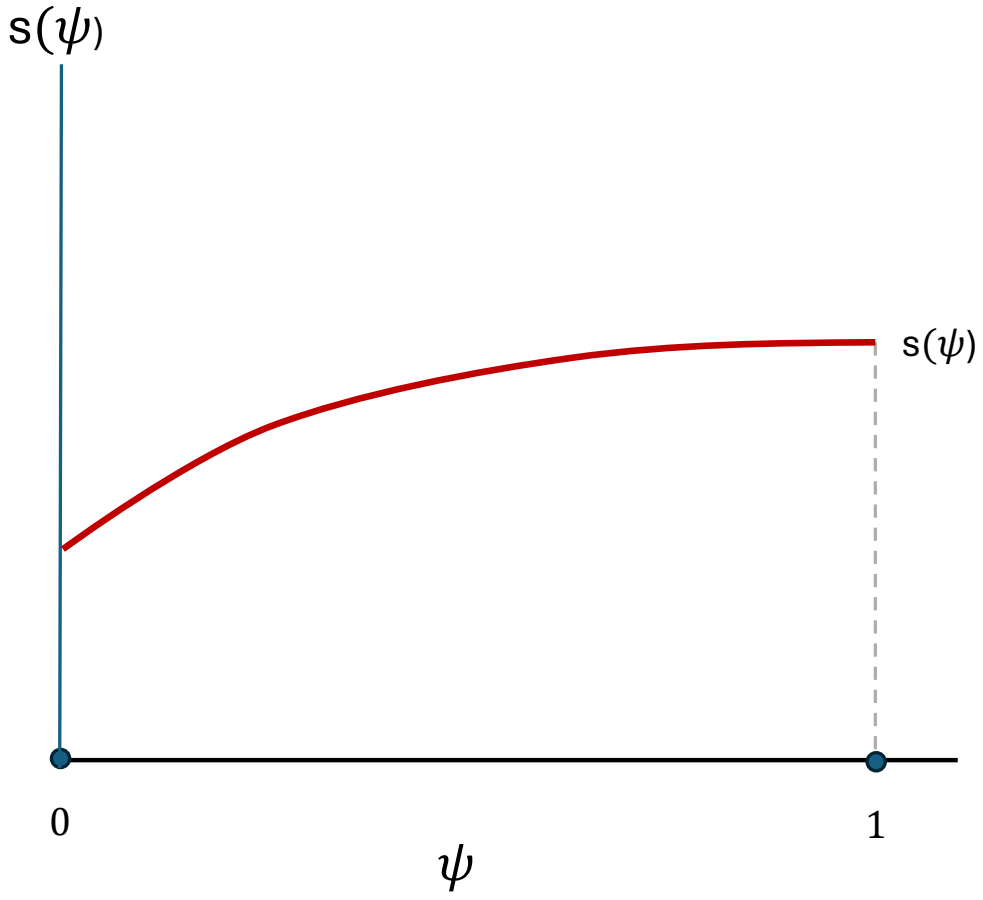
Relationship Endurance

- The probability of an enduring relationship, $p(e)$

$$p(e) = \phi(c(\psi), s(\psi))$$

  - Cohesion, $c(\psi)$
  - Spark, $s(\psi)$
  - Cultural distance, $\psi \in [0, 1]$

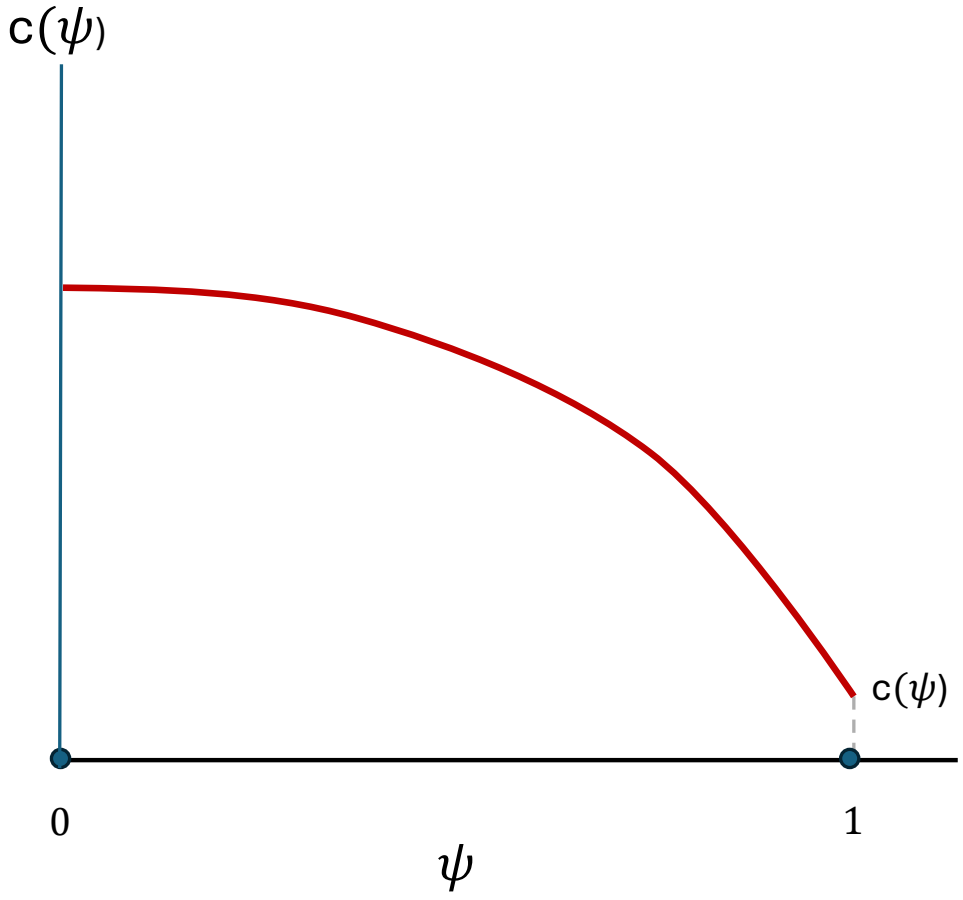
Spark & Cultural Distance



Cultural Distance

Mathematical Conditions – 1

Cohesion & Cultural Distance

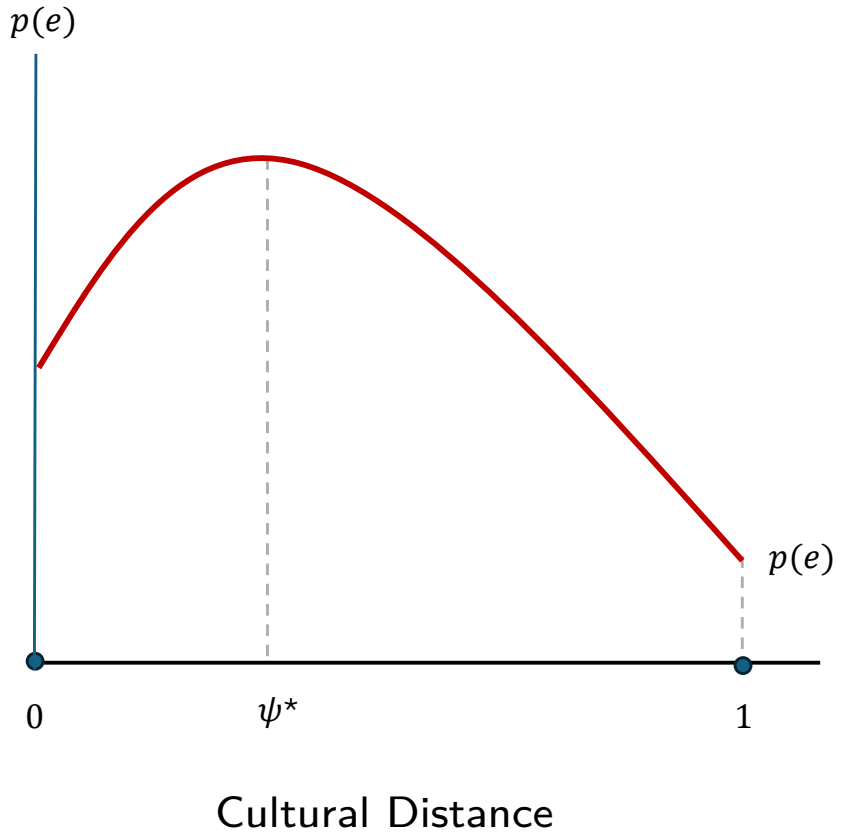


Cultural Distance

Mathematical Conditions – 2

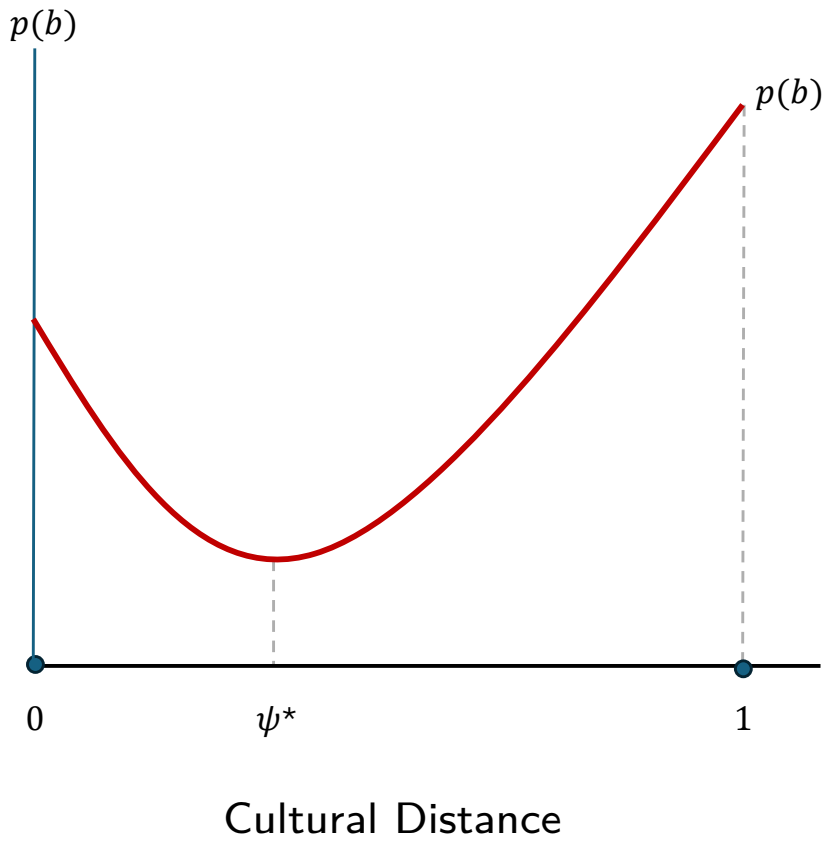
Cultural Distance & Probability of Enduring Relationship

- An intermediate level of cultural distance, ψ^*, maximizes the probability of an enduring relationship, $p(e)$



Cultural Distance & Probability of Breakup

- An intermediate level of cultural distance, ψ^*, minimizes the probability of a breakup, $p(b)$



Estimating the Effect of Cultural Distance

- Explore the impact of *ancestral cultural distance* on separation rates
  - Among couples born and residing in the US
    - Exposed to similar institutions & norms that could affect separation
  - Differ in their ancestral cultural backgrounds
    - Differ in their cultural synergy

Empirical Analysis Accounts For

- Ancestry fixed effects
  - All observable and unobservable factors associated with an ancestry that may be conducive for higher separation rates
    - short-term orientation; gender bias; diminished trust; limited cooperation; individualism
- Age-pair fixed effects
  - Impact of partners' ages & age gap on separation rates
- Education-pair fixed effects
  - Impact of partners' education & education gap on separation rates
- Continent-pair fixed effects
  - All observable and unobservable factors associated with the ancestral continent pair that may affect the separation rate

Main Variables and Data Sources

- Separation rates of nearly 1 million couples, reflecting over 2000 ancestry pairs
  - Current Population Survey (CPS): 1994–2024

    A monthly survey conducted by the U.S. Census Bureau and the Bureau of Labor Statistics to collect comprehensive data on labor force characteristics, employment, and demographic information. Widely regarded as highly reliable due to its rigorous sampling methods
- Dependent variable
  - Likelihood of separation *within a year*
- Independent variable
  - Cultural distance between the couple's ancestries

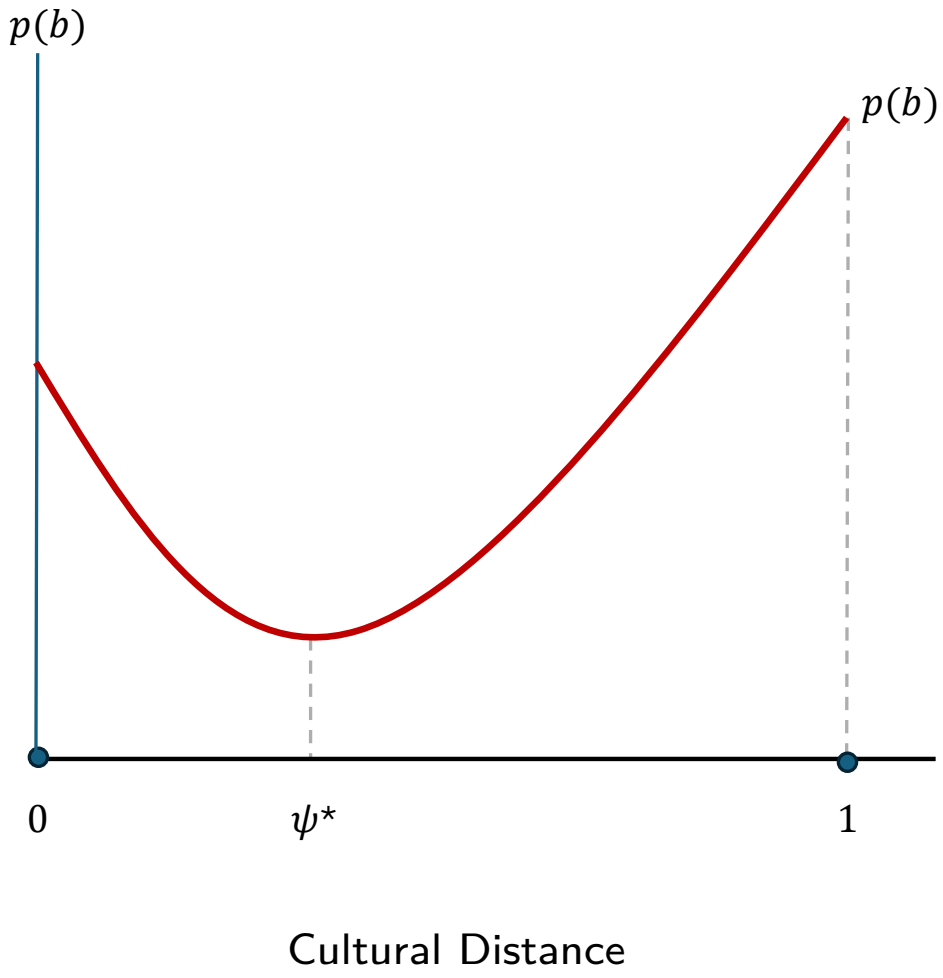
Empirical Model

$$B_{c,i,j,t} = \alpha + \beta_1 \psi_{ij} + \beta_2 \psi_{ij}^2 + \gamma_t + \gamma_{i,j} + \gamma_{ij} + \gamma_c + \epsilon_c$$

- $B_{c,t,i,j} \equiv \begin{cases} 1 & \text{if couple } c \text{ from ancestries } i,j \text{ separated within 1 year at time } t \\ 0 & \text{otherwise} \end{cases}$
- $\psi_{ij} \equiv$ cultural distance between the couple's ancestries i and j
- $\gamma_{i,j} \equiv$ ancestry fixed-effects
- $\gamma_{ij} \equiv$ ancestral continent pair fixed-effects
- $\gamma_c \equiv$ couple's age pair, education pair, and cohort pair fixed-effects
- $\gamma_t \equiv$ survey year t's fixed-effects
  - Hypothesis: $\beta_1 < 0$ and $\beta_2 > 0$

External Validity

The U-Shaped Effect of Cultural Distance on the Probability of Separation



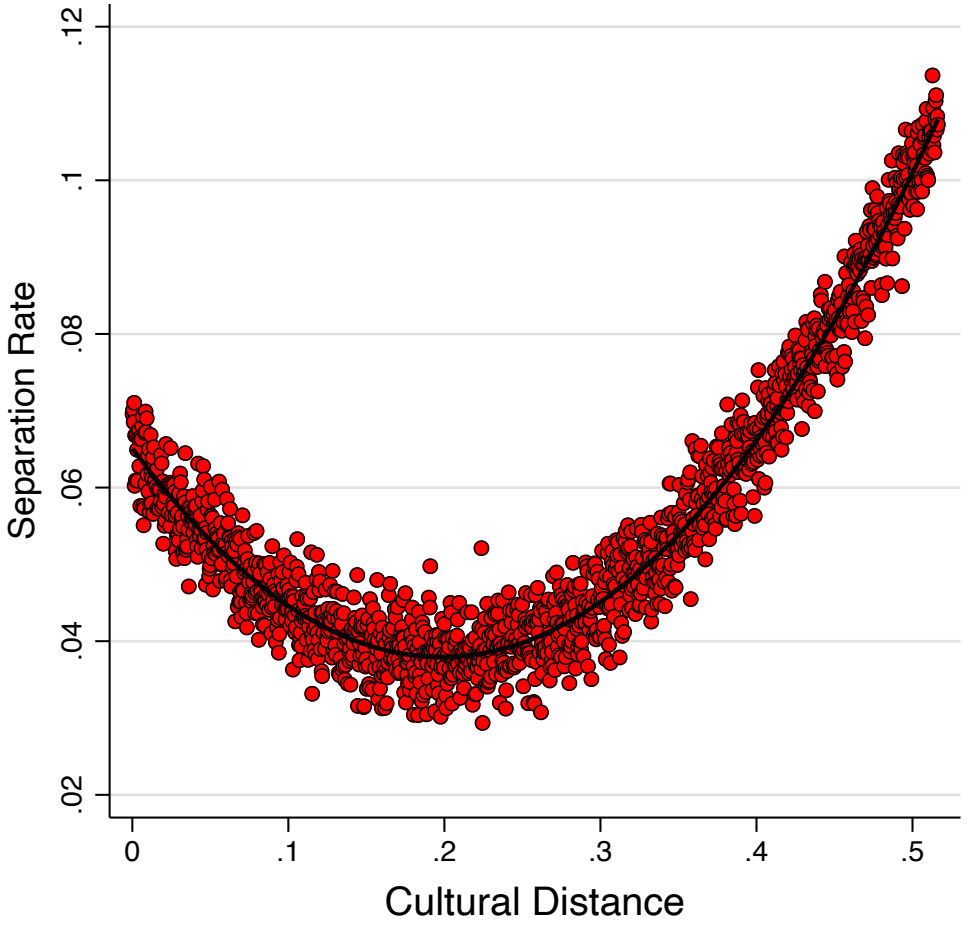
Cultural Distance and Separation Rate: Baseline Analysis

	Separation Rate		
	(1)	(2)	(3)
Cultural distance	−0.27***	−0.27***	−0.24***
	(0.062)	(0.064)	(0.062)
Cultural distance (squared)	0.67***	0.66***	0.62***
	(0.20)	(0.19)	(0.18)
Ancestry FE	✓	✓	✓
Ancestral Continent Pair FE	✓	✓	✓
Age Pair FE		✓	✓
Education Pair FE			✓
Observations	18,187	18,187	18,187
Ancestry pairs	2119	2119	2119
R^2	0.13	0.19	0.20
Optimal cultural distance	.199	.202	.195
Statistical significance (p-value $\leq$)	.004	.002	.003
Effect size ($\psi_{min} \rightarrow \psi^*$)	-41%	-41%	-37%
Effect size ($\psi_{max} \rightarrow \psi^*$)	-61%	-60%	-57%

All regressions include: (i) year fixed effects and (ii) cohort fixed effects

Education Local Effects Alternative Sample F_{ST}

Cultural Distance and Separation Rate: Baseline Analysis

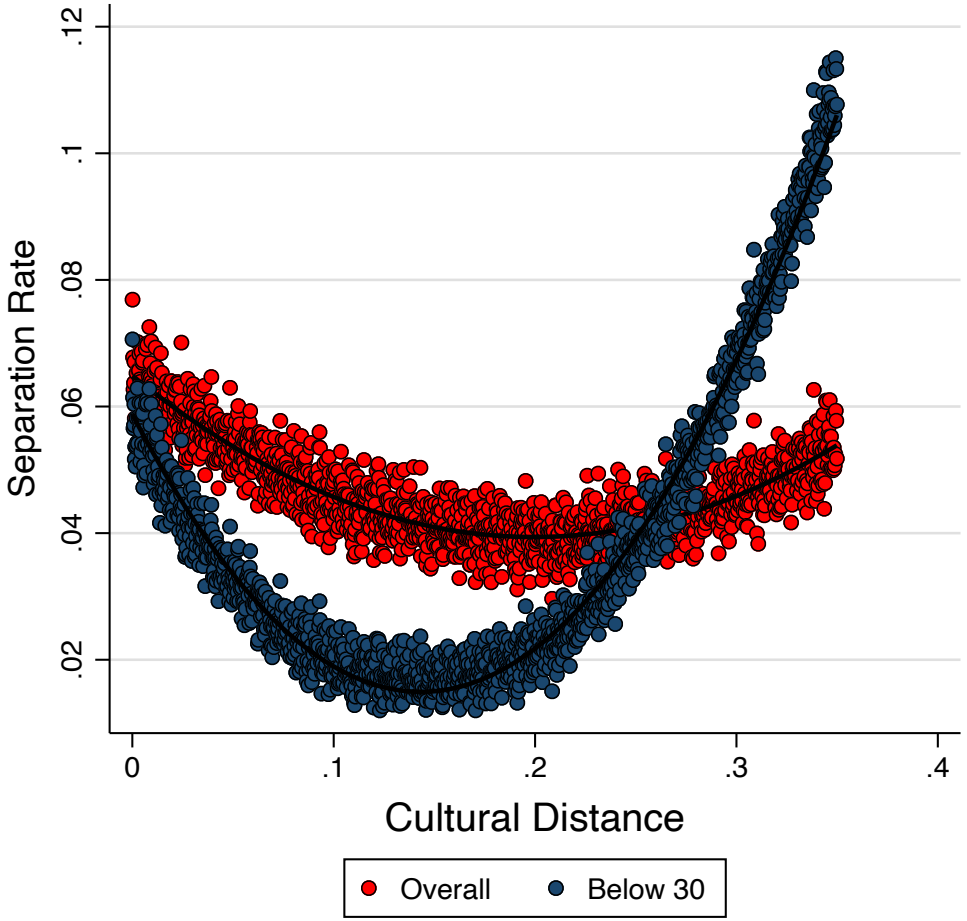


Cultural Distance and Separation Rate: Age Below 30

	Separation Rate		
	(1)	(2)	(3)
Cultural distance	−0.69***	−0.61***	−0.58***
	(0.19)	(0.19)	(0.19)
Cultural distance (squared)	2.40***	2.21***	2.08***
	(0.60)	(0.62)	(0.63)
Ancestry FE	✓	✓	✓
Ancestral Continent Pair FE	✓	✓	✓
Age Pair FE		✓	✓
Education Pair FE			✓
Observations	993	993	993
Ancestry pairs	203	203	203
R^2	0.24	0.31	0.32
Optimal cultural distance	.144	.138	.140
Statistical significance (p-value $\leq$)	.001	.001	.002
Effect size ($\psi_{min} \rightarrow \psi^*$)	-85%	-74%	-70%
Effect size ($\psi_{max} \rightarrow \psi^*$)	-96%	-93%	-92%

All regressions include: (i) year fixed effects and (ii) cohort fixed effects

Cultural Distance and Separation Rate: The Age Factor



The Algorithm

- Designed to enhance the user's matching process
  - Ranks potential matches based on their predicted *endurance*
  - Generates a *Relationship Synergy Score* on a 0–100 scale

Reprium's Seamless Integration into Existing Dating Platforms

- Minimal required user inputs:
  - Age (already available)
  - Education (already available)
  - Ancestry (users would select up to 2 ancestries from a menu of ancestral homelands)

Instantaneous computation of the Synergy Score

- Evaluating the synergy between Elizabeth and Johann:
  - Elizabeth:
    - 25 years old
    - College degree completed
    - British ancestry
  - Johann:
    - 27 years old
    - PhD
    - German-Polish ancestry

Instantaneous computation of the Synergy Score

- **Step 1:** Compute the couple's cultural distance
  - $CD_{(E,GP)}$ – retrieved from Reprium's cultural distance matrix
  - $CD^2_{(E,GP)}$ – square of the cultural distance value $CD_{(E,GP)}$
- **Step 2:** Compute the effect of $CD_{(E,GP)}$ and $CD^2_{(E,GP)}$ on their synergy score
  - Use the regression's estimated coefficients $coef_{cd}$ & $coef_{cd^2}$

$$CD_{\text{effect}} = coef_{cd} \times CD_{(E,GP)}$$

$$CD^2_{\text{effect}} = coef_{cd^2} \times CD^2_{(E,GP)}$$

Instantaneous computation of the Synergy Score

- **Step 3:** Retrieve fixed effects from the regression model
  - Age-pair fixed effect: $FE_{\text{age}}[25, 27]$
  - Education-pair fixed effect: $FE_{\text{edu}}[\text{College}, \text{PhD}]$
  - Ancestry fixed : $FE_{\text{ancestry}}[\text{British}] + FE_{\text{ancestry}}[\text{German-Polish}]$
  - Continent-pair fixed effect: $FE_{\text{continent}}[\text{Europe}, \text{Europe}]$
  - Constant from the regression model
- **Step 4:** Compute the predicted separation rate (by summing all components)

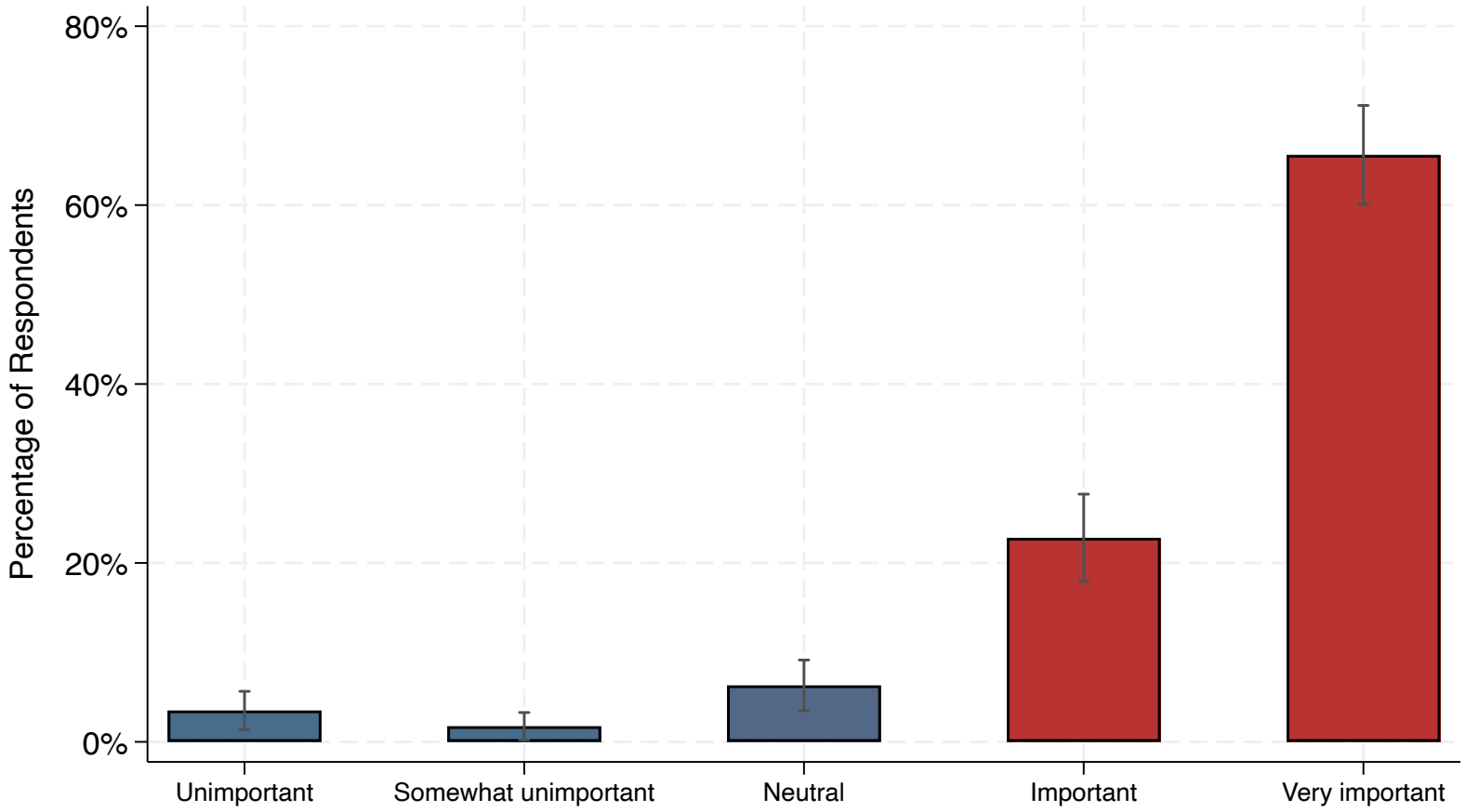
$$\widehat{B} = \text{Constant} + CD_{eff} + CD_{eff}^2 + FE_{\text{age}}[25, 27] + FE_{\text{edu}}[\text{College}, \text{PhD}] + FE_{\text{anc}}[\text{British}] + FE_{\text{anc}}[\text{German-Polish}] + FE_{\text{cont}}[\text{Europe}, \text{Europe}]$$

- **Step 5:** Obtain the *Relationship Synergy Score*
  - The algorithm ranks $\widehat{B}$ relative to all other potential matches, providing a score in the scale of 0 to 100

Child Well-Being

- Prosperity
- Education
- Creativity
- Focus

Importance of Child Well-Being



Survey question: "When choosing a partner, how important is it for you the well-being of your potential joint children?"

Survey conducted for Reprium by Prolific on a representative sample of the US population

Unfulfilled Demand

- Existing dating apps:
  - Ignore the impact of a match on the well-being of potential children
    - Focusing on relationship *formation* & the couple's well-being
  - Lack the technology to predict:
    - The well-being of prospective children that may emerge from the match

Reprium

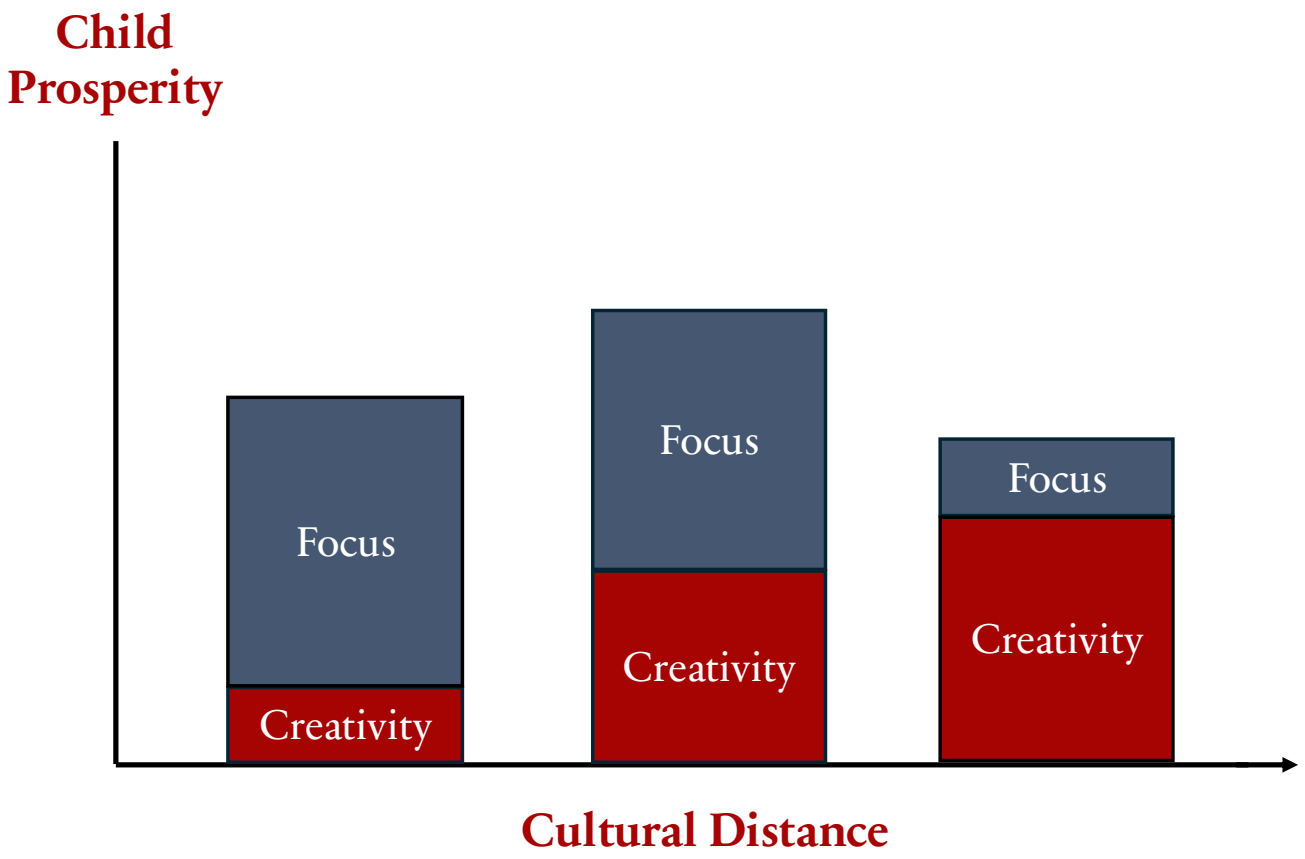
- Scientifically-based algorithm fostering the well-being of prospective children
- Providing a *Child Well-Being Score*
  - Education score
  - Creativity score
  - Attention span score
  - Earning Capacity score
- Scores reflect:
  - The couple's cultural distance
  - Ancestral predisposition toward prosperity, education, creativity & focus:

    (i) long-term orientation; (ii) individualism; (iii) obedience
  - Education-pair

Cultural Synergy: The Key to Child Well-Being

- Children of culturally similar parents tend to be:
  - More focused but less creative
- Children of highly dissimilar parents then to be:
  - More creative but less focused
- An intermediate level of cultural dissimilarity strikes the ideal balance
  - Sufficient differences for creativity & similarities for sustained focus

Co



Theoretical Model

Estimating the Effect of Cultural Synergy

- Explore the impact of *parental cultural distance* on a child well-being
  - Among individuas born and reside in the US
    - Exposed to similar incentives & education system
  - Differ in their ancestral cultural backgrounds
    - Differ in their parental cultural synergy

Empirical Analysis Accounting For

- Parental ancestry fixed effects
  - All observable and unobservable factors associated with an ancestry that may be conducive for education, creativity, prosperity, and focus

    (long-term orientation, individualism, obdience)
- Education-pair fixed effects
  - Impact of parental education & education gap on the child well-being
- Continent-pair fixed effects
  - All observable and unobservable factors associated with the ancestral continent pair that may affect the child well-being
- Age and sex fixed effects

Main Variables & Data Sources

- Income of nearly 4 million individuals, originating from 3,000 ancestry pairs
  - Data Sources:
    - American Community Survey (ACS), 2001-2023

      The ACS is an annual survey conducted by the U.S. Census Bureau that collects detailed data on demographics, economics, housing, and social characteristics across the country. It provides a reliable and representative snapshot of the U.S. population, widely used for policymaking, research, and planning
    - Censuses – 1990 & 2000

      The U.S. Census is a nationwide count of the population conducted every ten years by the U.S. Census Bureau, as mandated by the Constitution. It provides essential data for congressional representation, federal funding allocation, and demographic analysis
    - Sample: US-born, full-time year-round workers, employed in the private sector, age 25-64
- Dependent variable:
  - Individual's wage income
- Independent variable:
  - Cultural distance between parental ancestries

The Empirical Model

$$\ln w_{k,i,j,t} = \alpha + \beta_1 \psi_{ij} + \beta_2 \psi_{ij}^2 + \gamma_t + \gamma_{i,j} + \gamma_{ij} + \gamma_k + \epsilon_k$$

- $w_{k,t,i,j} \equiv$ wage of individual k, from ancestries i & j, surveyed at year t
- $\psi_{ij} \equiv$ cultural distance between ancestries i & j
- $\gamma_{i,j} \equiv$ ancestry fixed-effects
- $\gamma_{ij} \equiv$ ancestral continent pair fixed-effects
- $\gamma_k \equiv$ vector of individual-level fixed-effects: age, cohort, gender
- $\gamma_t \equiv$ survey year t's fixed effects
  - Hypothesis: $\beta_1 > 0$ and $\beta_2 < 0$

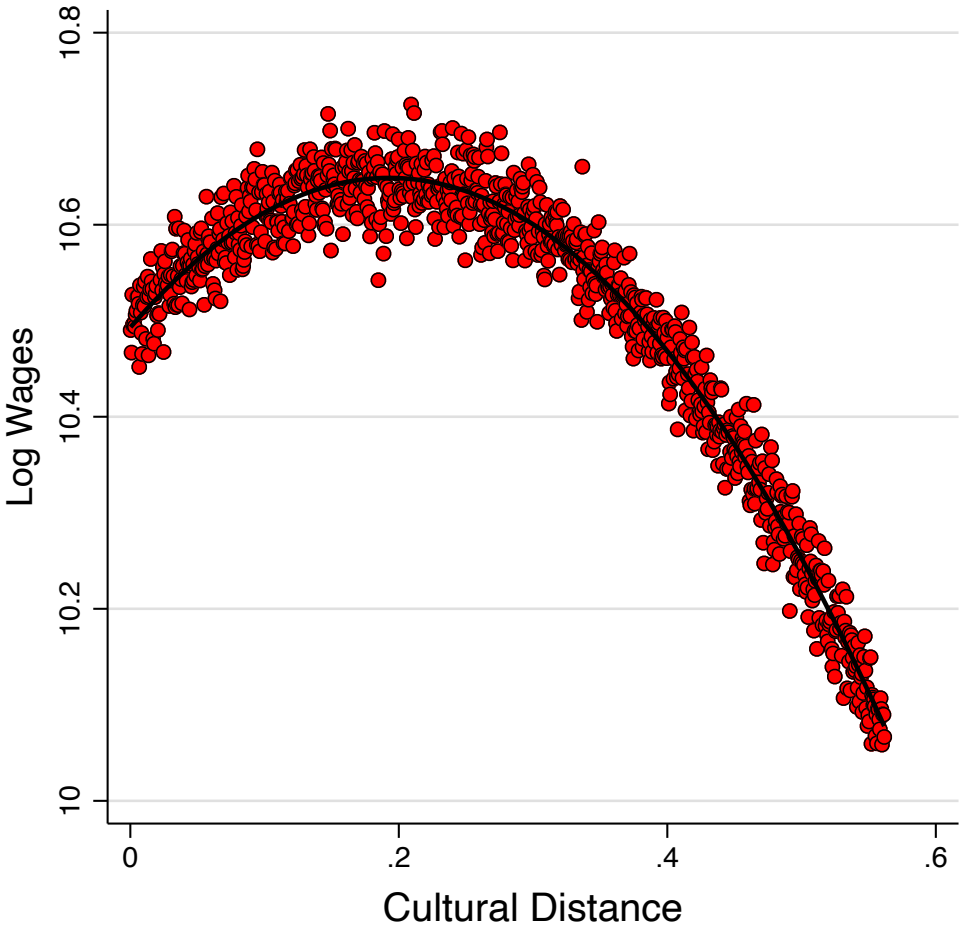
Couple's Cultural Distance and Child Wage Income: Baseline Analysis

	Log wages		
	(1)	(2)	(3)
Cultural distance	1.68***	1.67***	1.63***
	(0.44)	(0.44)	(0.44)
Cultural distance (squared)	−4.39***	−4.38***	−4.23***
	(0.89)	(0.89)	(0.85)
Ancestry FE	✓	✓	✓
Ancestral Continent Pair FE	✓	✓	✓
Age FE		✓	✓
Sex FE			✓
Observations	3,727,398	3,727,398	3,727,398
Ancestry pairs	3403	3403	3403
R^2	0.058	0.070	0.12
Optimal cultural distance	.190	.191	.192
Statistical significance (p-value $\leq$)	.001	.001	.001
Effect size ($\psi_{min} \rightarrow \psi^*$)	17%	17%	17%
Effect size ($\psi_{max} \rightarrow \psi^*$)	83%	83%	78%

All regressions include: (i) year fixed effects and (ii) cohort fixed effects

Age Groups | Part Time | Decades | Educ & Occup FE | Local Analysis | Education | NLSY | NLSY-Edu | Fst

Couple's Cultural Distance and Child Prosperity



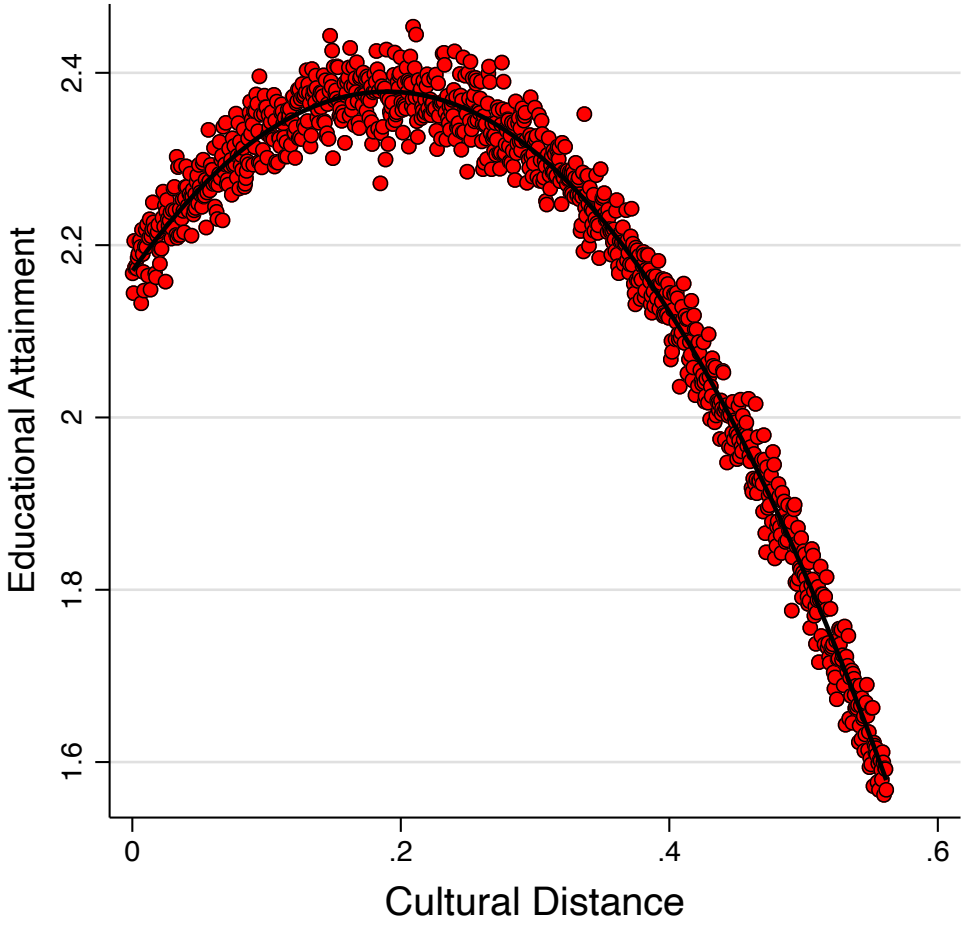
Cultural Distance & Child's Well-Being Beyond Prosperity

- The impact of cultural distance on various child characteristics
  - Educational attainment
  - Creativity
  - Attention span (Focus)

Couple's Cultural Distance & Child's Educational Attianment

	Educational Attainment		
	(1)	(2)	(3)
Cultural distance	2.21***	2.20***	2.21***
	(0.81)	(0.81)	(0.81)
Cultural distance (squared)	−5.82***	−5.81***	−5.82***
	(1.54)	(1.54)	(1.54)
Ancestry FE	✓	✓	✓
Ancestral Continent Pair FE	✓	✓	✓
Age FE		✓	✓
Sex FE			✓
Observations	3,727,398	3,727,398	3,727,398
Ancestry pairs	3403	3403	3403
R^2	0.060	0.061	0.061
Optimal cultural distance	.189	.189	.189
Statistical significance (p-value $\leq$)	.004	.004	.004

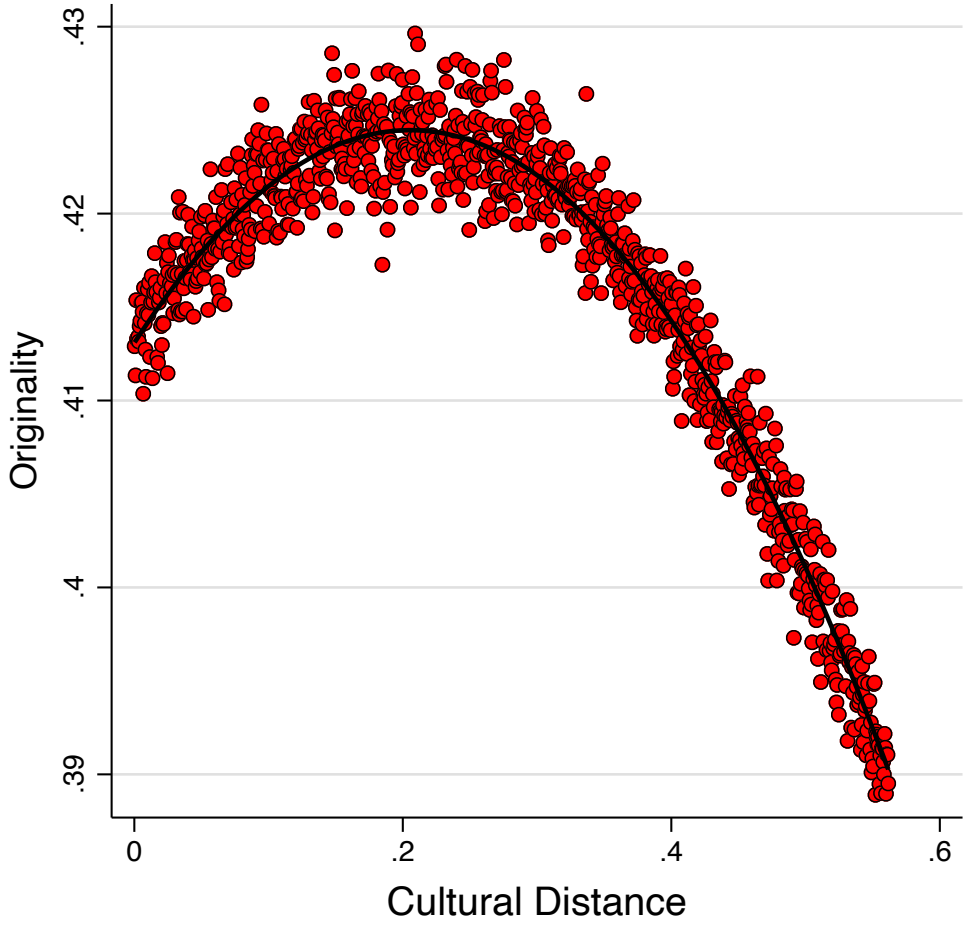
Couple's Cultural Distance & Child Educational Attainment



Couple's Cultural Distance and Child's Cognitive Flexibility

	Originality		
	(1)	(2)	(3)
Cultural distance	0.12***	0.11***	0.11***
	(0.032)	(0.031)	(0.032)
Cultural distance (squared)	−0.28***	−0.28***	−0.27***
	(0.062)	(0.062)	(0.060)
Ancestry FE	✓	✓	✓
Ancestral Continent Pair FE	✓	✓	✓
Age FE		✓	✓
Sex FE			✓
Observations	1,784,760	1,784,760	1,784,760
Ancestry pairs	2928	2928	2928
R^2	0.013	0.016	0.027
Optimal cultural distance	.202	.202	.205
Statistical significance (p-value $\leq$)	.001	.001	.001

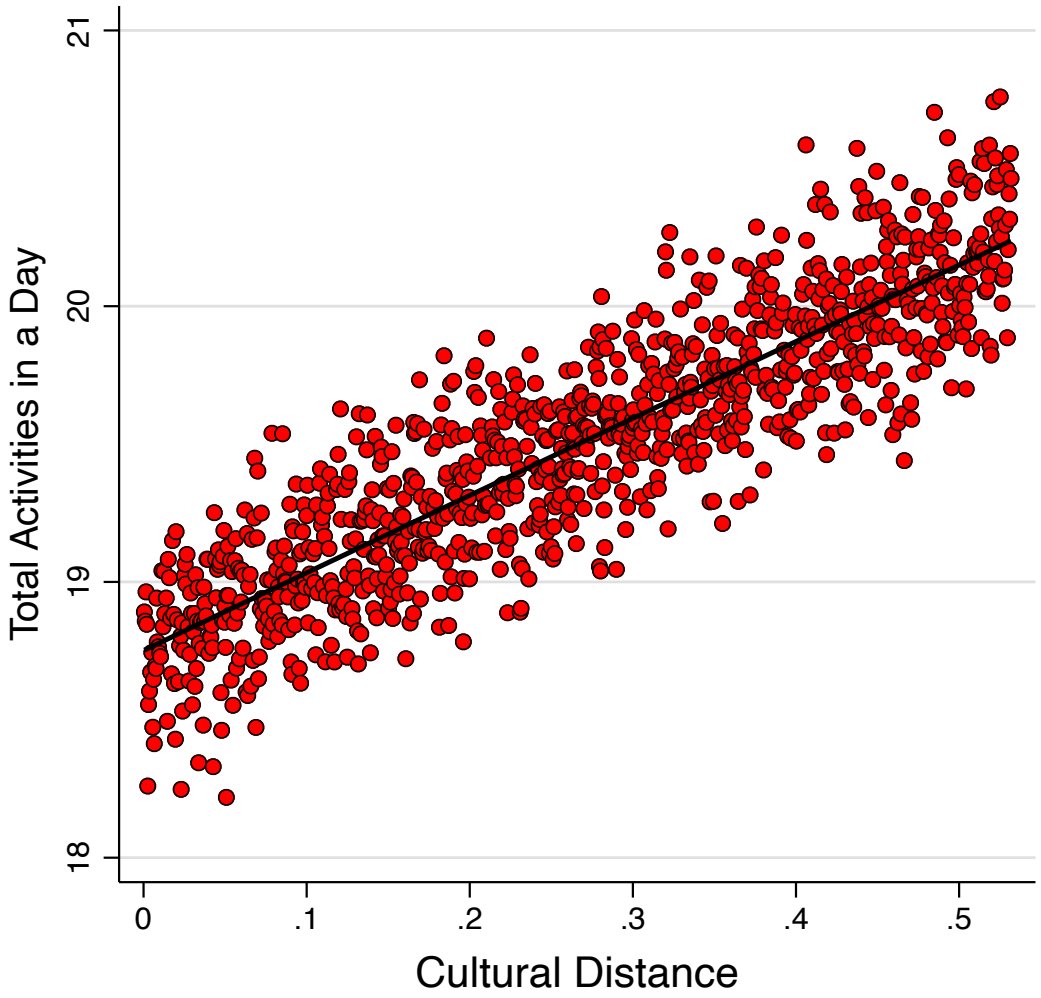
Couple's Cultural Distance & Child's Originality



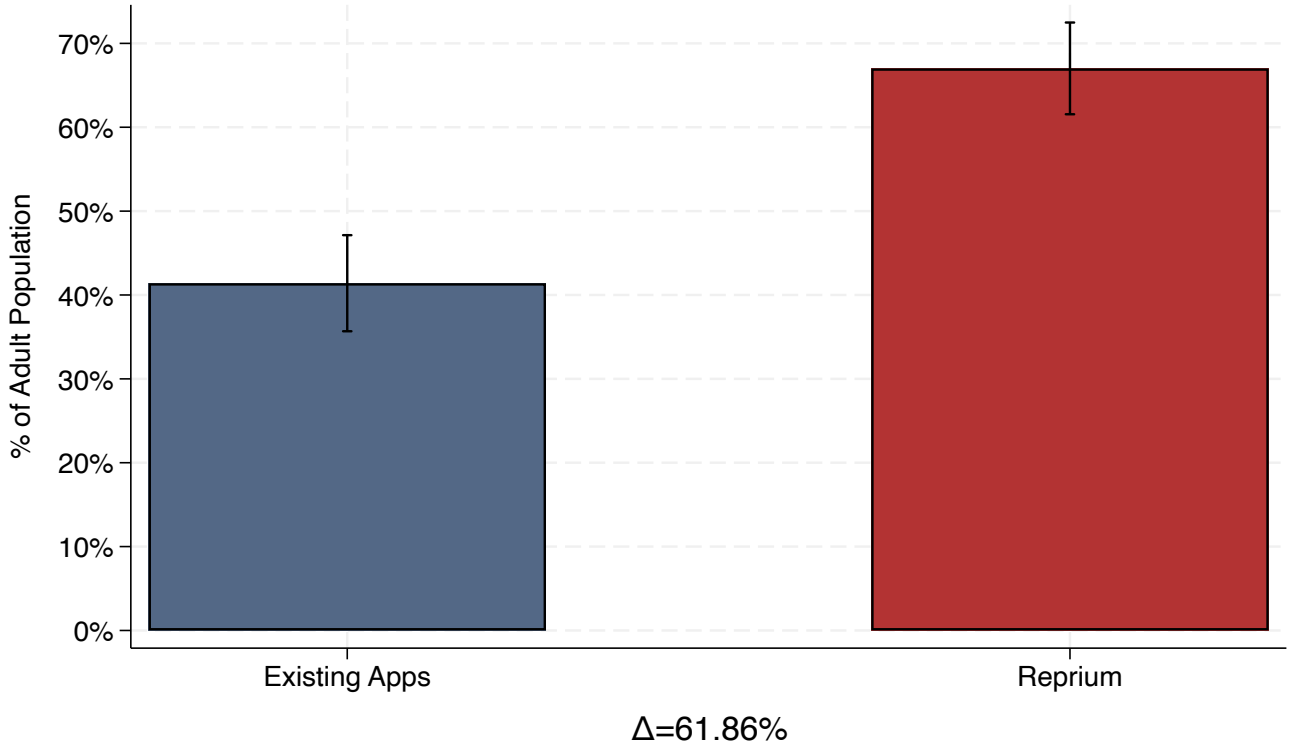
Couple's Cultural Distance & Child's Lack of Focus

	Total activities in a day		
	(1)	(2)	(3)
Cultural distance	2.77***	2.70***	2.86***
	(1.02)	(1.03)	(0.94)
Ancestry FE	✓	✓	✓
Ancestral Continent Pair FE	✓	✓	✓
Age FE		✓	✓
Sex FE			✓
Observations	9,368	9,368	9,368
Ancestry pairs	589	589	589
R^2	0.079	0.098	0.12

Couple's Cultural Distance & Child's Lack of Focus



Expansion of US User Base Driven by Integration of R's Features

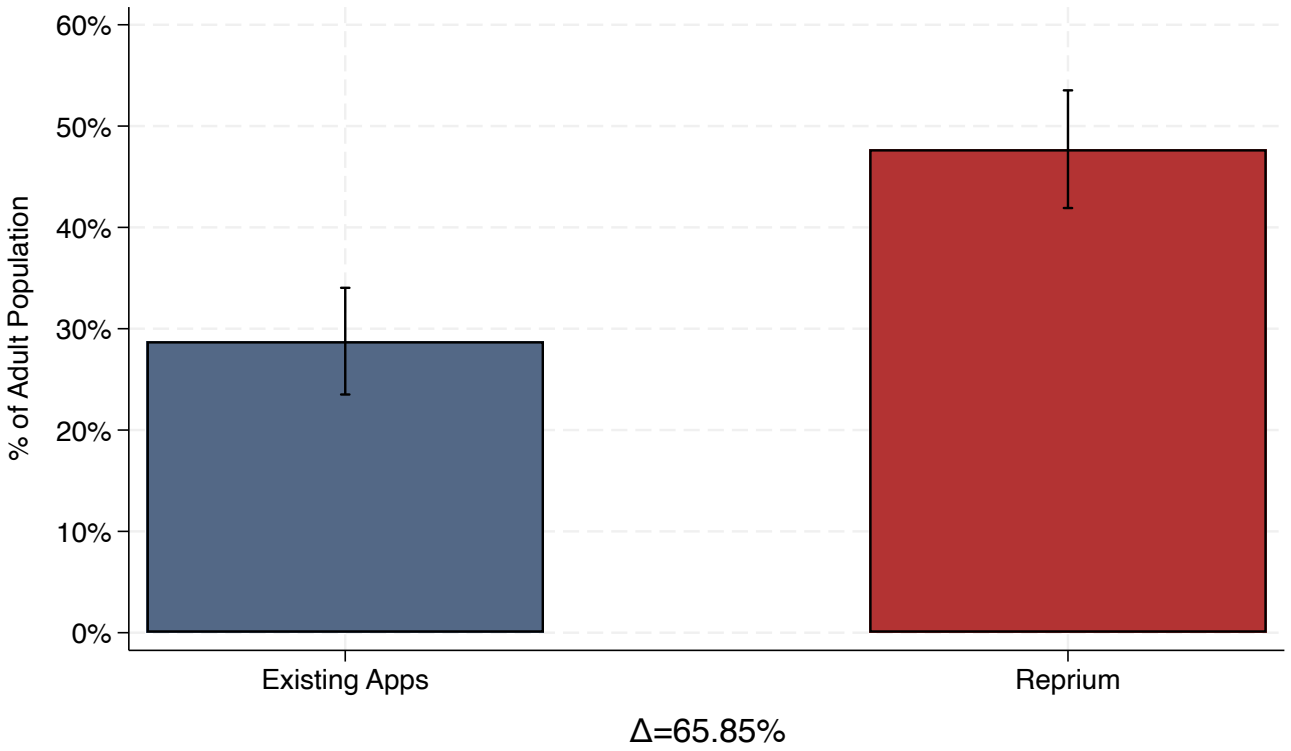


Survey conducted for *Reprium* by *Prolific* on a representative sample of the US population

41% of survey participants would use a dating app to search for partners—comparable to 37% of the adult U.S. population who have actually used a dating app

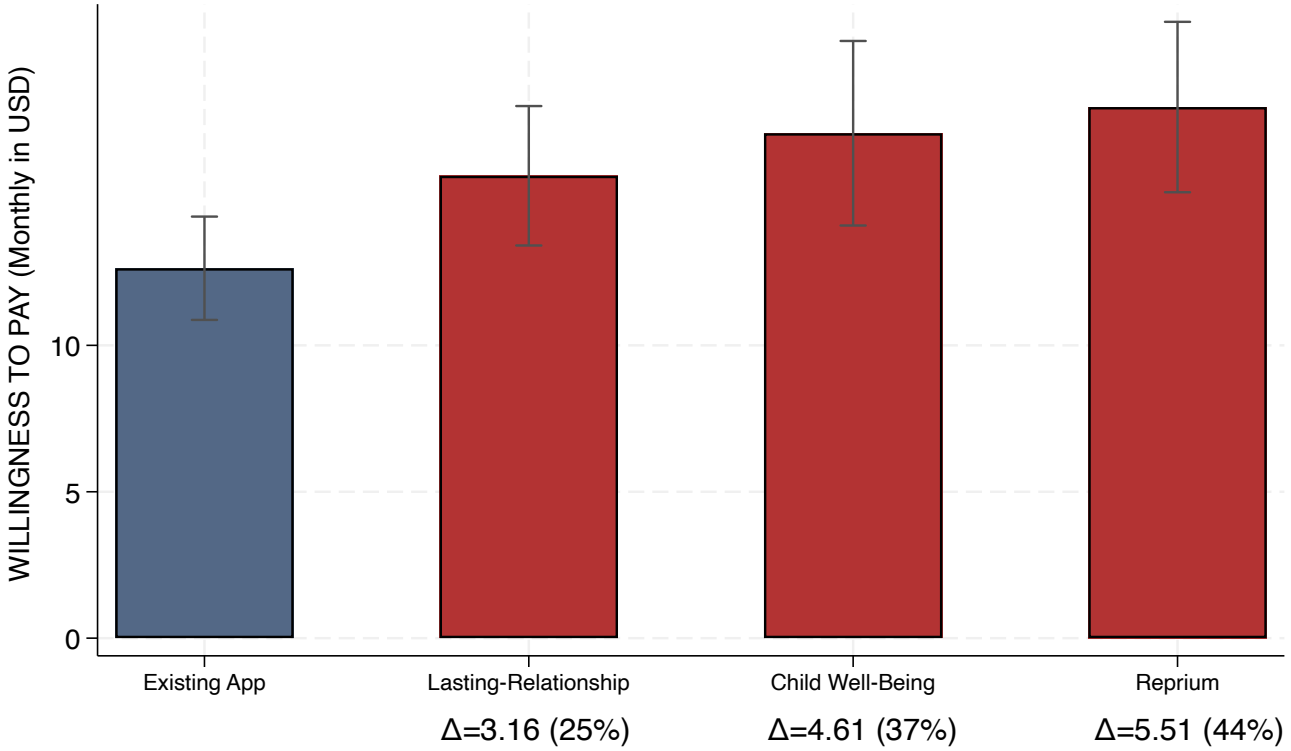
Survey Question: If you are not using a dating app when looking for a partner, would you use a dating app if it offers cutting-edge, scientifically validated features that accurately predict the likelihood of (i) having a long-lasting relationship and (ii) having a joint child with a high level of well-being?

Expansion of US Paid User Base Driven by Integration of R's Features



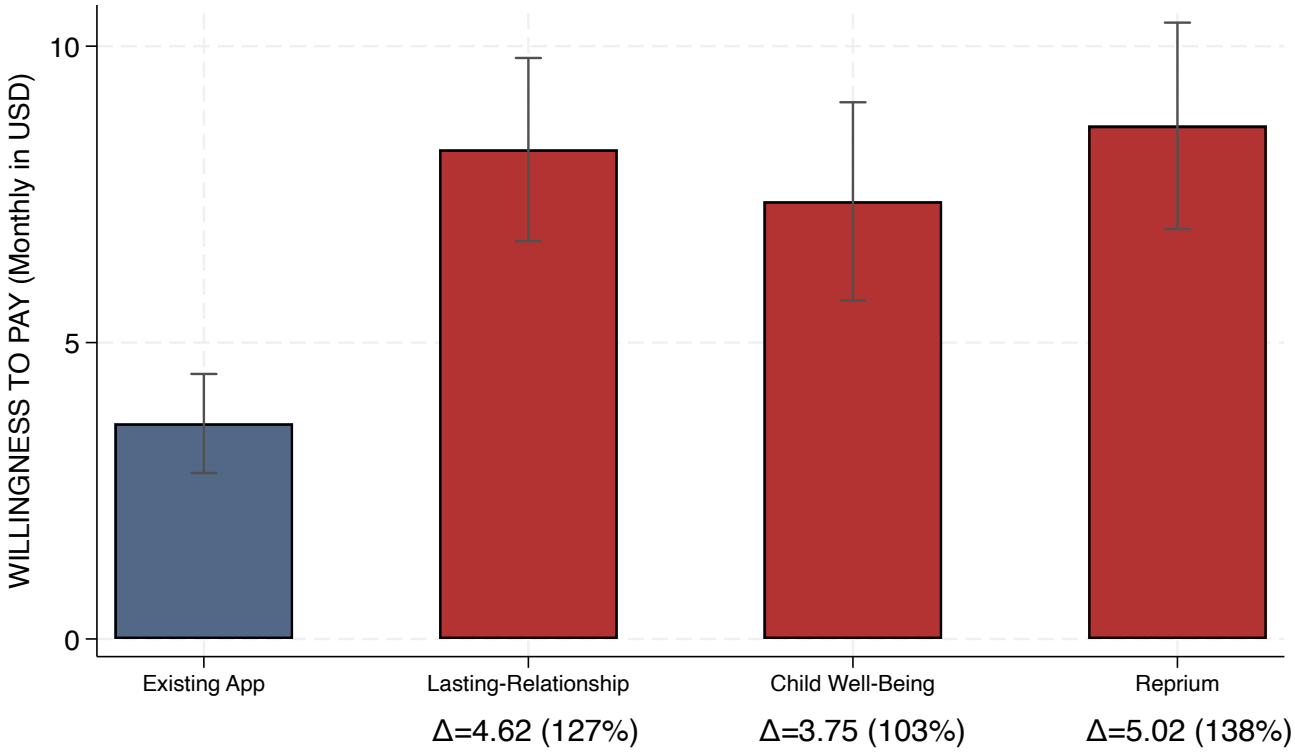
Survey conducted for *Reprium* by *Prolific* on a representative sample of the US population

Boost in Willingness to Pay Driven by Integration of R's Features



Survey conducted for *Reprium* by *Prolific* on a representative sample of the US population

Industry Revenue Boost Driven by Integration of R's Features



Survey conducted for Reprium by *Prolific* on a representative sample of the US population.

The average monthly willingness to pay in our survey is $3.64—comparable to the actual $3.54 paid by Match Group's U.S. users and the $4.88 average across all dating app users in the U.S

Survey Question: Suppose you are looking for a partner. How much are you willing to pay per month for your favorite dating app? How much extra are you willing to pay per month for your favorite dating app if it would add cutting-edge, scientifically validated features that accurately predict the likelihood of (i) having a long-lasting relationship, (ii) having a joint child with a high level of well-being, (iii) both?

Reprium – Game-Changing Target of Acquisition

- If Bumble integrates R's novel features:
  - It would capture the entire industry-wide expansion of 66% in paid user
  - Paid user base would grow by 4.07x (based 21.5% US user share)
  - Payment per paid user would rise by 44%
  - US Annual revenues would surge 5.86x (from $0.61B to $3.57B)
  - Europe's Annual revenue would surge (from $0.21B to $1.23B) (assuming comparable expansion in user base and payment per user)
  - Annual revenue from the US & Europe would grow by 3.99B

Reprium

Science-Based Fortune Teller

Empirical Design

- Reverse causality — not feasible
  - A couple's separation rate cannot affect their ancestral cultural distance
- Omitted variables
  - Unobserved heterogeneity across ancestral origins
    - Accounted for by ancestry fixed effects
  - Unobserved heterogeneity across ancestry pairs
    - Accounted for by cultural distance
    - Controlled for using continent-pair fixed effects

Ω

Couple's Cultural Distance – Measurement

- Couple's Cultural Distance
  - ⇒ Cultural disparities between the couple's ancestral origins
- Measurement: (reflecting decade of research in cultural economics)
  - Pre-historic migratory distance between the couple's ancestries
  - Accounting for Post-1500 migration flows & multicultural societies

    Cultural distance between Argentina & Bolivia is measured as the weighted average of cultural distances between:

    % of Argentinians originating from Italy & % of Bolivians originating from Spain

    % of Native Argentinians & % of Bolivians originating from Spain

    % of Argentinians originating from Italy & % of Native Bolivians

    % of Native Argentinians & % of Native Bolivians
- Dominates alternative measures:
  - Fst Genetic Distances

Ω

Separation Rate: Married & Unmarried Couples

- Probability of separation

$$P(s) = \Phi(\psi, \psi^2, \text{age-pair}_{fe}, \text{educ-pair}_{fe}, \text{ancestry}_{fe}, M)$$

- Married couples ($M = 1$) have lower separation than unmarried ($M = 0$)

$$P(s)_{M=1} < P(s)_{M=0}$$

- Being married reduces the separation rate by a *constant* but has little if any effect on the impact of cultural distance on separation rate

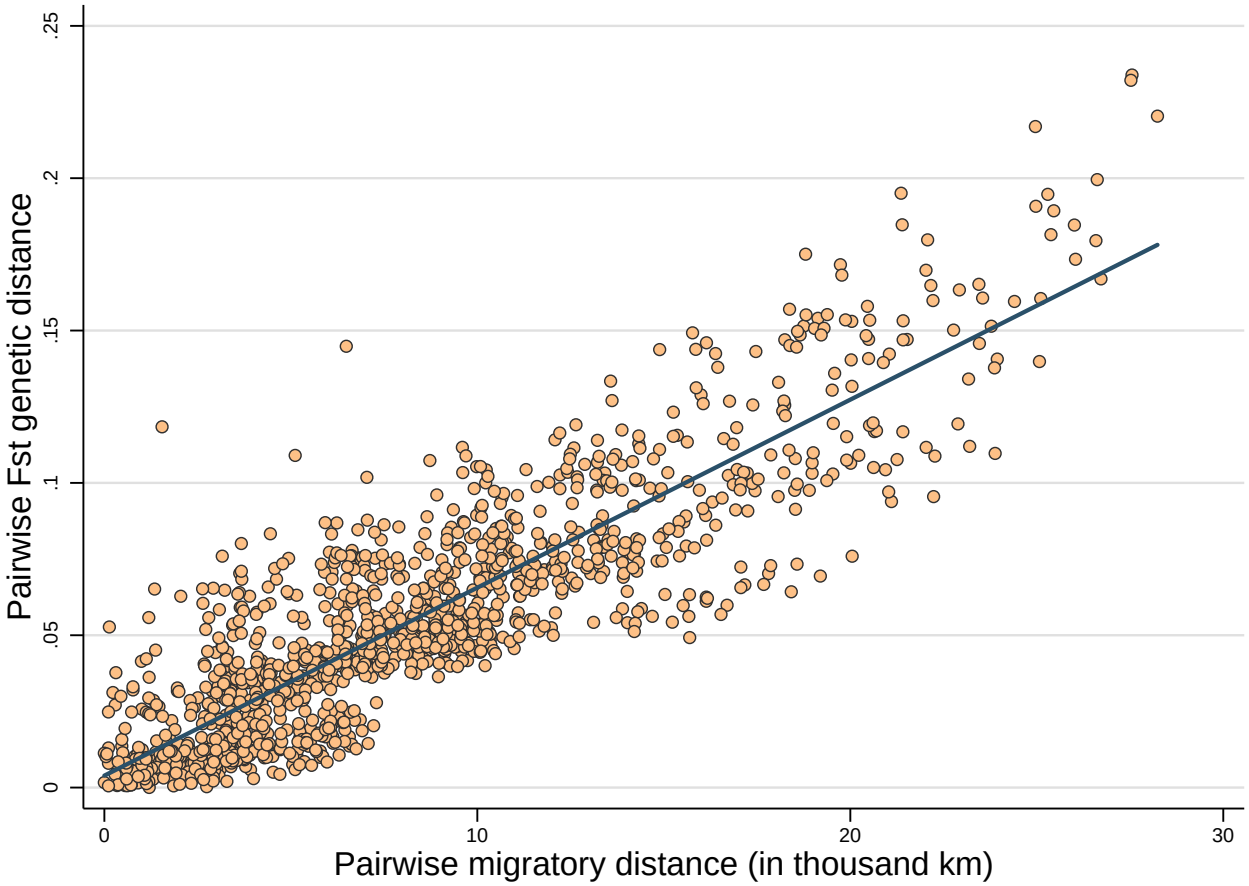
$$\left.\frac{\partial P(s)}{\partial \psi}\right|_{M=0} \approx \left.\frac{\partial P(s)}{\partial \psi}\right|_{M=1} \qquad \left.\frac{\partial^2 P(s)}{\partial \psi^2}\right|_{M=0} \approx \left.\frac{\partial^2 P(s)}{\partial \psi^2}\right|_{M=1}$$

  - Although US regions differ in their divorce probability they have no effect on the impact of cultural distance on separation

- Being married reduces the separation rate by a *constant* but has no affect the ranking produced by our algorithm

Local Effects Ω

Migratory Distances & Fst Genetic Distances



Cultural Distance and Separation Rate Across Educational Categories

	Separation Rate			
	HS Completed		College Graduates	
	(1)	(2)	(3)	(4)
Cultural distance	−0.28**	−0.25**	−0.31***	−0.30***
	(0.11)	(0.12)	(0.10)	(0.10)
Cultural distance (squared)	0.77***	0.69**	0.89***	0.79***
	(0.29)	(0.33)	(0.34)	(0.26)
Ancestry FE	✓	✓	✓	✓
Ancestral Continent Pair FE	✓	✓	✓	✓
Age Pair FE		✓		✓
Observations	10,892	10,892	3,055	3,055
Ancestry pairs	1216	1216	822	822
R^2	0.16	0.24	0.26	0.40
Optimal cultural distance	.180	.178	.176	.187
Statistical significance (p-value $\leq$)	.007	.024	.012	.004
Effect size ($\psi_{min} \rightarrow \psi^*$)	-35%	-31%	-59%	-59%
Effect size ($\psi_{max} \rightarrow \psi^*$)	-60%	-56%	-83%	-80%

All regressions include: (i) year fixed effects and (ii) cohort fixed effects

Ω

Cultural Distance and Separation Rate: Local Effects

	Separation Rate			
	(1)	(2)	(3)	(4)
Cultural distance	−0.24***	−0.24***	−0.24***	−0.24**
	(0.062)	(0.063)	(0.064)	(0.11)
Cultural distance (squared)	0.62***	0.61***	0.60***	0.65**
	(0.18)	(0.19)	(0.19)	(0.28)
Region FE		✓		
State FE			✓	
Micro area FE				✓
Observations	18,187	18,187	18,187	9,514
Ancestry pairs	2119	2119	2119	1429
R^2	0.20	0.20	0.20	0.28
Optimal cultural distance	.195	.197	.197	.189
Statistical significance (p-value $\leq$)	.002	.004	.004	.015
Effect size ($\psi_{min} \rightarrow \psi^*$)	-37%	-37%	-37%	-37%
Effect size ($\psi_{max} \rightarrow \psi^*$)	-57%	-57%	-57%	-60%

All regressions include: (i) year fixed effects and (ii) cohort fixed effects

Ω

Cultural Distance and Separation Rate: Alternative Sample

	Separation Rate		
	(1)	(2)	(3)
Cultural distance	−0.17***	−0.15***	−0.14***
	(0.028)	(0.028)	(0.028)
Cultural distance (squared)	0.36***	0.31***	0.29***
	(0.10)	(0.10)	(0.10)
Ancestry FE	✓	✓	✓
Ancestral Continent Pair FE	✓	✓	✓
Age Pair FE		✓	✓
Education Pair FE			✓
Observations	836,762	836,762	836,762
Ancestry pairs	2369	2369	2369
R^2	0.046	0.065	0.066
Optimal cultural distance	.237	.248	.244
Statistical significance (p-value $\leq$)	.008	.023	.025
Effect size ($\psi_{min} \rightarrow \psi^*$)	-47%	-45%	-40%
Effect size ($\psi_{max} \rightarrow \psi^*$)	-50%	-42%	-40%

All regressions include: (i) year fixed effects and (ii) cohort fixed effects

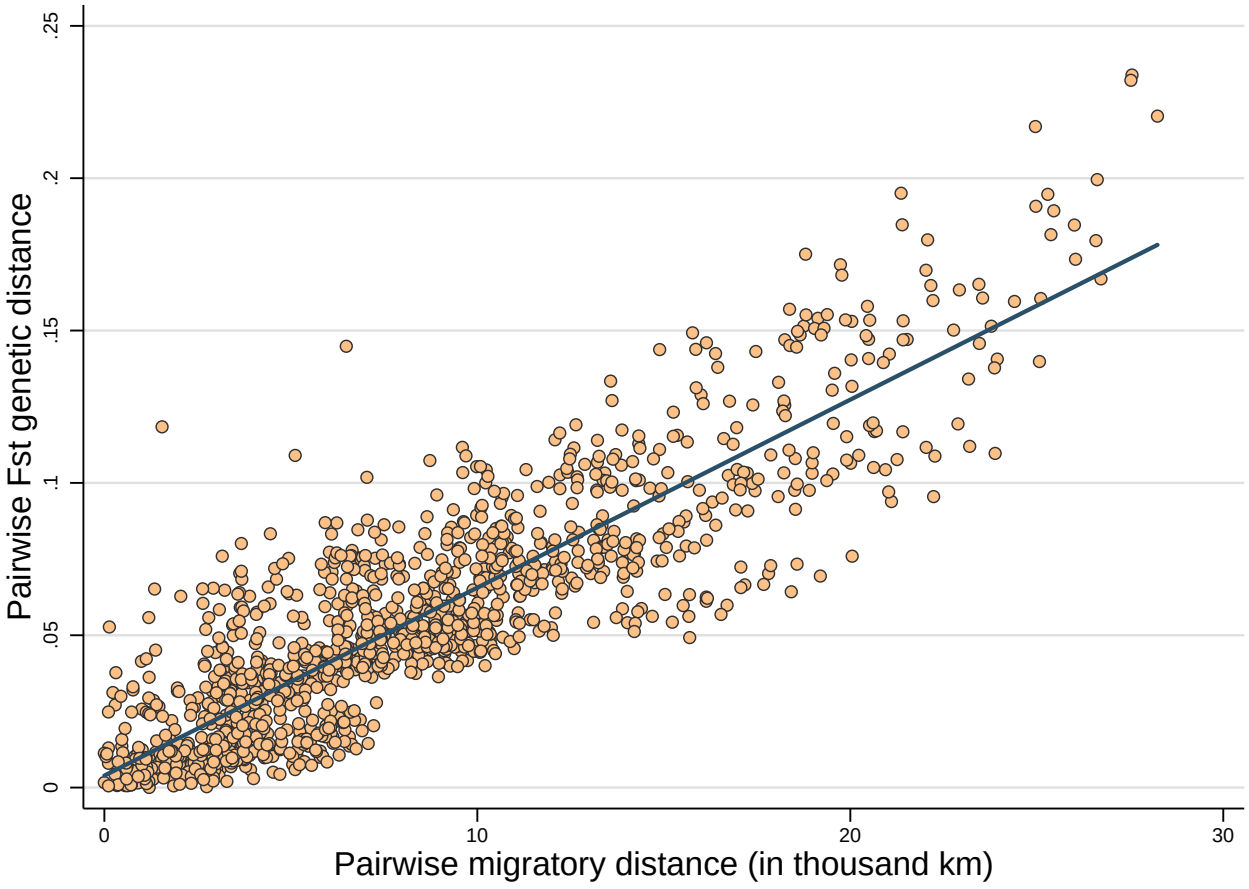
Ω

Cultural Distance and Separation Rate: Fst Genetic Distance

	Separation Rate		
	(1)	(2)	(3)
Cultural distance	−0.23***		−0.23***
	(0.072)		(0.067)
Cultural distance (squared)	0.57**		0.51**
	(0.22)		(0.24)
Genetic distance		−0.13**	−0.011
		(0.066)	(0.068)
Genetic distance (squared)		0.36**	0.12
		(0.15)	(0.16)
Observations	17,831	17,831	17,831
Ancestry pairs	2045	2045	2045
R^2	0.20	0.20	0.20
Cultural distance:			
Optimal distance	.197		.220
Statistical significance (p-value $\leq$)	.015		.062
Effect size ($\psi_{min} \rightarrow \psi^*$)	-35%		-38%
Effect size ($\psi_{max} \rightarrow \psi^*$)	-55%		-49%
Genetic distance:			
Optimal distance		.187	.046
Statistical significance (p-value $\leq$)		.021	.434
Effect size ($Fst_{min} \rightarrow Fst^*$)		-20%	0%
Effect size ($Fst_{max} \rightarrow Fst^*$)		-74%	-55%

All regressions include: (i) year FE; (ii) Cohort FE; (iii) Ancestry FE; (iv) Continent Pair FE; (v) Age FE; (vi) Sex FE

Migratory Distances & Fst Genetic Distances



Cultural Distance & Wages: Estimates by Age Groups

	Log wages			
	25-34	35-44	45-54	55-64
	(1)	(2)	(3)	(4)
Cultural distance	1.23***	1.78***	1.82***	1.72***
	(0.36)	(0.48)	(0.49)	(0.49)
Cultural distance (squared)	−3.21***	−4.65***	−4.66***	−4.43***
	(0.68)	(0.96)	(0.96)	(1.14)
Observations	1,044,380	1,040,509	991,973	650,511
Ancestry pairs	2910	2465	2040	1548
R^2	0.093	0.080	0.099	0.11
Optimal cultural distance	.191	.191	.195	.194
Statistical significance (p-value $\leq$)	.001	.001	.001	.001

All regressions include: (i) year FE; (ii) Cohort FE; (iii) Ancestry FE; (iv) Continent Pair FE; (v) Age FE; (vi) Sex FE

Ω

Cultural Distance & Wages: Including Non Full-Time Workers

	Log wages		
	(1)	(2)	(3)
Cultural distance	1.63***	1.63***	1.57***
	(0.44)	(0.44)	(0.42)
Cultural distance (squared)	−4.43***	−4.42***	−4.21***
	(0.92)	(0.92)	(0.86)
Ancestry FE	✓	✓	✓
Ancestral Continent Pair FE	✓	✓	✓
Age FE		✓	✓
Sex FE			✓
Observations	4,957,998	4,957,998	4,957,998
Ancestry pairs	3623	3623	3623
R^2	0.037	0.047	0.10
Optimal cultural distance	.184	.184	.186
Statistical significance (p-value $\leq$)	.001	.001	.001

All regressions include: (i) year fixed effects and (ii) cohort fixed effects

Cultural Distance & Wage Income: Estimates per Decade

	Log wages			
	1990	2000	2010	2020
	(1)	(2)	(3)	(4)
Cultural distance	1.84***	1.74***	1.54***	1.55***
	(0.63)	(0.50)	(0.42)	(0.35)
Cultural distance (squared)	−4.19***	−4.37***	−4.10***	−4.45***
	(1.33)	(1.01)	(0.83)	(0.74)
Observations	590,909	1,414,725	1,239,205	482,540
Ancestry pairs	1318	2273	2827	2506
R^2	0.17	0.12	0.12	0.093
Optimal cultural distance	.219	.199	.188	.174
Statistical significance (p-value $\leq$)	.002	.001	.001	.001

All regressions include: (i) year FE; (ii) Cohort FE; (iii) Ancestry FE; (iv) Continent Pair FE; (v) Age FE; (vi) Sex FE

Ω

Cultural Distance & Wages: Within Educational & Occupational Categories

	Log wages			
	(1)	(2)	(3)	(4)
Cultural distance	1.63***	0.91***	1.04***	0.78***
	(0.44)	(0.21)	(0.25)	(0.18)
Cultural distance (squared)	−4.23***	−2.37***	−2.69***	−2.03***
	(0.85)	(0.43)	(0.49)	(0.36)
Education FE		✓		✓
Occupation FE			✓	✓
Observations	3,727,398	3,727,398	3,727,395	3,727,395
Ancestry pairs	3403	3403	3403	3403
R^2	0.12	0.32	0.34	0.41
Optimal cultural distance	.192	.193	.193	.192
Statistical significance (p-value $\leq$)	.001	.001	.001	.001

All regressions include: (i) year FE; (ii) Cohort FE; (iii) Ancestry FE; (iv) Continent Pair FE; (v) Age FE; (vi) Sex FE

Ω

Cultural Distance & Wage Income: Local Analysis

	Log wages			
	(1)	(2)	(3)	(4)
Cultural distance	1.61***	1.36***	1.14***	0.63***
	(0.43)	(0.38)	(0.33)	(0.18)
Cultural distance (squared)	−4.23***	−3.51***	−2.80***	−1.47***
	(0.85)	(0.71)	(0.61)	(0.38)
Region FE		✓		
State FE			✓	
Micro area FE				✓
Observations	2,128,922	2,128,922	2,128,922	2,128,922
Ancestry pairs	3116	3116	3116	3116
R^2	0.12	0.13	0.15	0.23
Optimal cultural distance	.19	.192	.203	.213
Statistical significance (p-value $\leq$)	.001	.001	.001	.001

All regressions include: (i) year FE; (ii) Cohort FE; (iii) Ancestry FE; (iv) Continent Pair FE; (v) Age FE; (vi) Sex FE

Ω

Accounting for Additional Wage Determinants - Baseline

	Log wages			
	(1)	(2)	(3)	(4)
Cultural distance	4.64***			3.06**
	(0.72)			(1.08)
Cultural distance (squared)	−12.0***			−7.94**
	(1.97)			(2.90)
Cognitive ability		0.30***		0.26***
		(0.035)		(0.038)
Education (father)			0.045***	0.030**
			(0.011)	(0.012)
Education (mother)			0.026**	0.0014
			(0.012)	(0.018)
Observations	1,351	1,351	1,351	1,351
Individuals	555	555	555	555
Ancestry pairs	52	52	52	52
R^2	0.15	0.25	0.19	0.27
Optimal cultural distance	.193			.192
Statistical significance (p-value $\leq$)	.001			.017
Effect size ($\psi_{min} \rightarrow \psi^*$)	57%			34%
Effect size ($\psi_{max} \rightarrow \psi^*$)	105%			62%

Ω

Accounting for Additional Wage Determinants - College Graduates

	Log wages			
	(1)	(2)	(3)	(4)
Cultural distance	5.17			4.93
	(2.78)			(3.14)
Cultural distance (squared)	−17.2**			−17.2*
	(7.32)			(7.73)
Cognitive ability		0.13***		0.13**
		(0.027)		(0.052)
Education (father)			0.014	0.014
			(0.016)	(0.017)
Education (mother)			−0.018	−0.025
			(0.016)	(0.030)
Observations	523	523	523	523
Individuals	210	210	210	210
Ancestry pairs	34	34	34	34
R^2	0.21	0.22	0.21	0.23
Optimal cultural distance	.150			.143
Statistical significance (p-value $\leq$)	.050			.077
Effect size ($\psi_{min} \rightarrow \psi^*$)	48%			42%
Effect size ($\psi_{max} \rightarrow \psi^*$)	200%			219%

Couple's Cultural Distance & Child Wages: Fst Genetic Distance

	Log wages		
	(1)	(2)	(3)
Cultural distance	1.78***		1.59***
	(0.52)		(0.49)
Cultural distance (squared)	−4.56***		−4.18***
	(1.13)		(1.01)
Genetic distance		0.40***	0.34***
		(0.13)	(0.13)
Genetic distance (squared)		−0.26**	−0.23**
		(0.10)	(0.11)
Observations	3,010,196	3,010,196	3,010,196
Ancestry pairs	2776	2776	2776
R^2	0.12	0.12	0.12
Optimal cultural distance	.195		.190
Statistical significance (p-value $\leq$)	.001		.001

All regressions include: (i) year FE; (ii) Cohort FE; (iii) Ancestry FE; (iv) Continent Pair FE; (v) Age FE; (vi) Sex FE

Ω

Cultural Distance & Wages: Estimates By Education

	Log Wages			
	HS Completed		College Graduates	
	(1)	(2)	(3)	(4)
Cultural distance	0.91***	0.88***	1.32***	1.27***
	(0.27)	(0.25)	(0.32)	(0.31)
Cultural distance (squared)	−2.47***	−2.37***	−3.54***	−3.33***
	(0.64)	(0.58)	(0.60)	(0.57)
Ancestry FE	✓	✓	✓	✓
Ancestral Continent Pair FE	✓	✓	✓	✓
Age FE		✓		✓
Sex FE		✓		✓
Observations	2,336,623	2,336,623	1,390,773	1,390,773
Ancestry pairs	2654	2654	3041	3041
R^2	0.053	0.13	0.093	0.16
Optimal cultural distance	.184	.185	.186	.190
Statistical significance (p-value $\leq$)	.001	.001	.001	.001

All regressions include: (i) year FE; (ii) Cohort FE; (iii) Ancestry FE; (iv) Continent Pair FE; (v) Age FE; (vi) Sex FE

Ω

Spark and Cultural Distance

- The couple's spark, denoted by s

$$s = s(\psi)$$

- Spark increases with cultural distance, at a decreasing rate

$$s'(\psi) > 0 \qquad s''(\psi) < 0$$

- Boundary condition: the effect vanishes when cultural distance is maximal

$$s'(1) = 0$$

Cohesion and Cultural Distance

- The couple's cohesion, c

$$c = c(\psi)$$

- Cohesion decreases with cultural distance, at an increasing rate

(cultural similarities increases cohesion in decreasing rates)

$$c = c'(\psi) < 0 \qquad\qquad c = c''(\psi) < 0$$

- Boundary conditions

$$\lim_{\psi \to 0} c'(\psi) = 0 \quad \text{and} \quad c'(1) < 0$$

A Conceptual Framework

- Continuum of individuals that differ in their parental cultural dissimilarities

  $\rightarrow$ diversity of their mindset, ψ

$$\psi \in [0, 1]$$

  - Formative years: Acquire a range skills
  - Working years: Supply their human capital in the labor market

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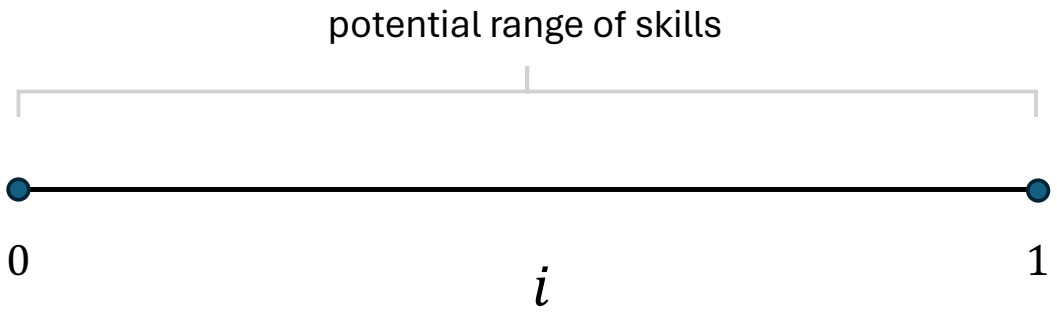
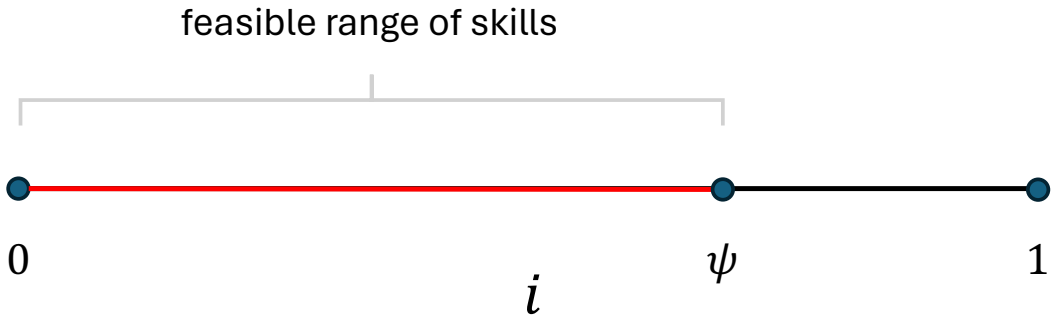
Potential Range of Skills

- The individual's stock of human capital is determined by:
  - Continuum of potential skills, $i \in [0, 1]$
    - There is no hierarchy of skills



Feasible Range of Skills

- The *feasible* range of productive skills that an individual can acquire, $i \in [0, \psi]$
  - Dictated by the diversity of the individual's mindset, $\psi \in [0, 1]$
    - More diverse mindset $\rightarrow$ greater feasible range of skills



Human Capital Formation

- Human capital acquired by individuals with diverse mindset, ψ

$$H(\psi) = \int_0^{\psi} s(i)^{\theta} di \quad \theta \in (0,1)$$

  - $s(i) \in (0,1) \equiv$ fraction of time devoted to the acquisition of skill i

Time Devoted to the Acquisition of Skills

- Time endowment:
  - Unit time endowment devoted to skill acquisition
- Time constraint in the acquisition of skills:

$$\int_0^{\psi} s(i)\,di \leq 1$$

  - $s(i) \in (0,1) \equiv$ fraction of time devoted to the acquisition of skill i
  - $[0,\psi] \equiv$ feasible range of acquired skills for individuals with diversity, ψ

Time Devoted to the Acquisition of Productive Skills

- If individuals allocate their time so as to maximize their human capital
  - Diminishing returns & symmetry in the effect of skills on HC $\rightarrow$

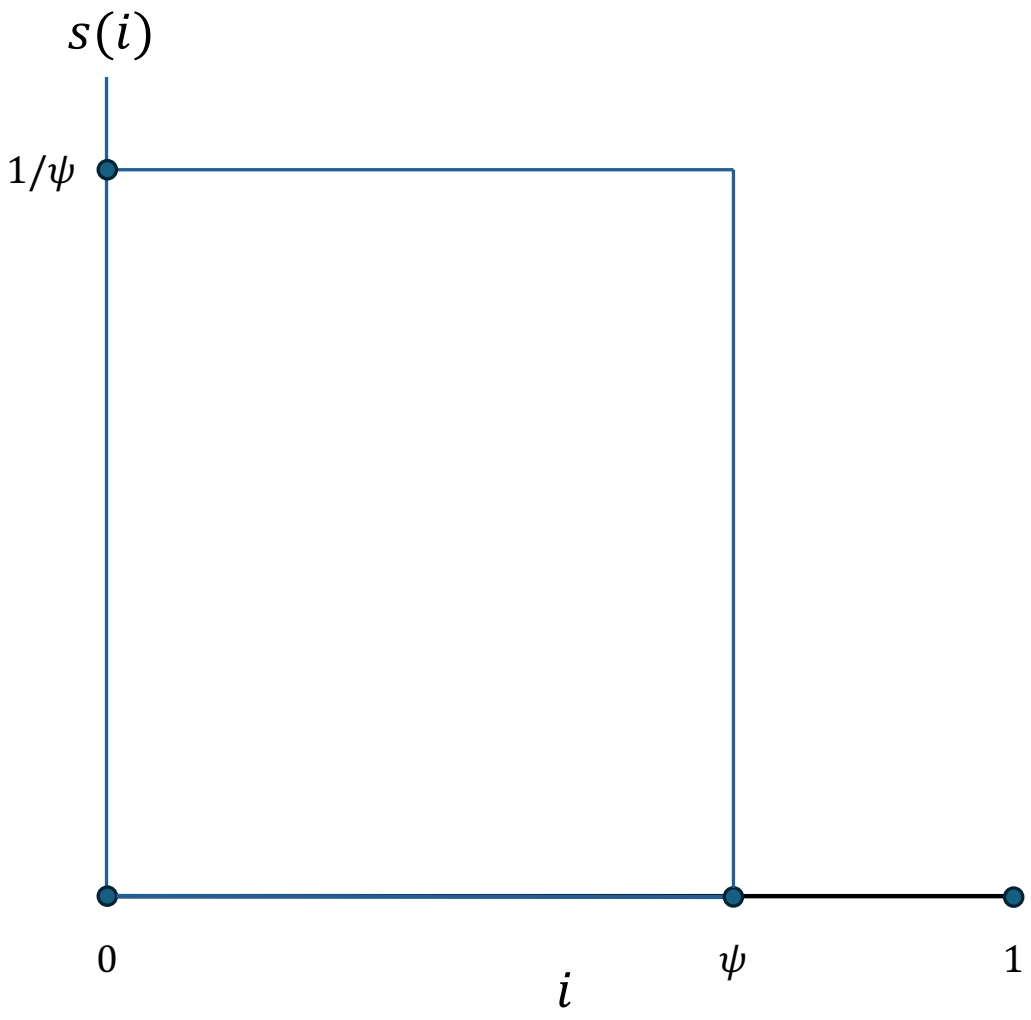
$$s(i) = s \quad \forall i \qquad \Rightarrow \qquad \psi s(i) = 1$$

  - The amount of skill i acquired by an individual with diversity ψ

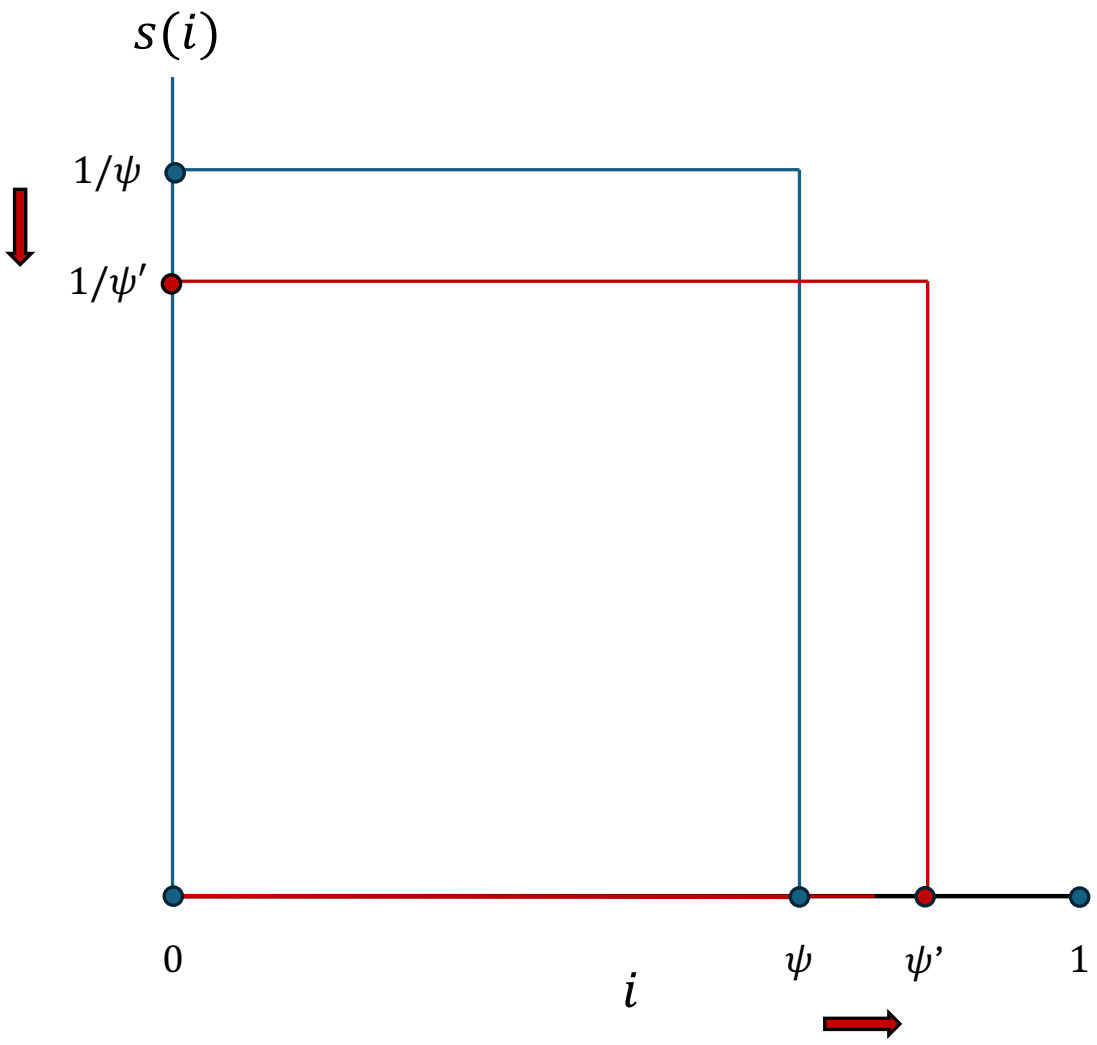
$$s(i) = 1/\psi \quad \forall i$$

    - Amount of each skill is inversely related to Cultural Distance

Time Devoted to the Acquisition of skill i, $s(i)$



Trade-off between Diversity and Depth



Individual's Stock of Human Capital

$$H(\psi) = \int_0^{\psi} s(i)^{\theta} di = \int_0^{\psi} (1/\psi)^{\theta} di = \psi^{1-\theta} \qquad \theta \in (0,1)$$

$$H'(\psi) > 0 \quad \& \quad H''(\psi) < 0$$

Individual's Effective Human Capital & Earning Capacity

- Diverse mindset $\Rightarrow$ diffused attention in any given task
- Fraction ψ of the stock of human capital is lost due to diffused attention
  - Effective human capital:

$$H^e(\psi) = (1-\psi)\int_0^{\psi} s(i)^{\theta} di = (1-\psi)\psi^{1-\theta}; \qquad \theta \in (0,1)$$

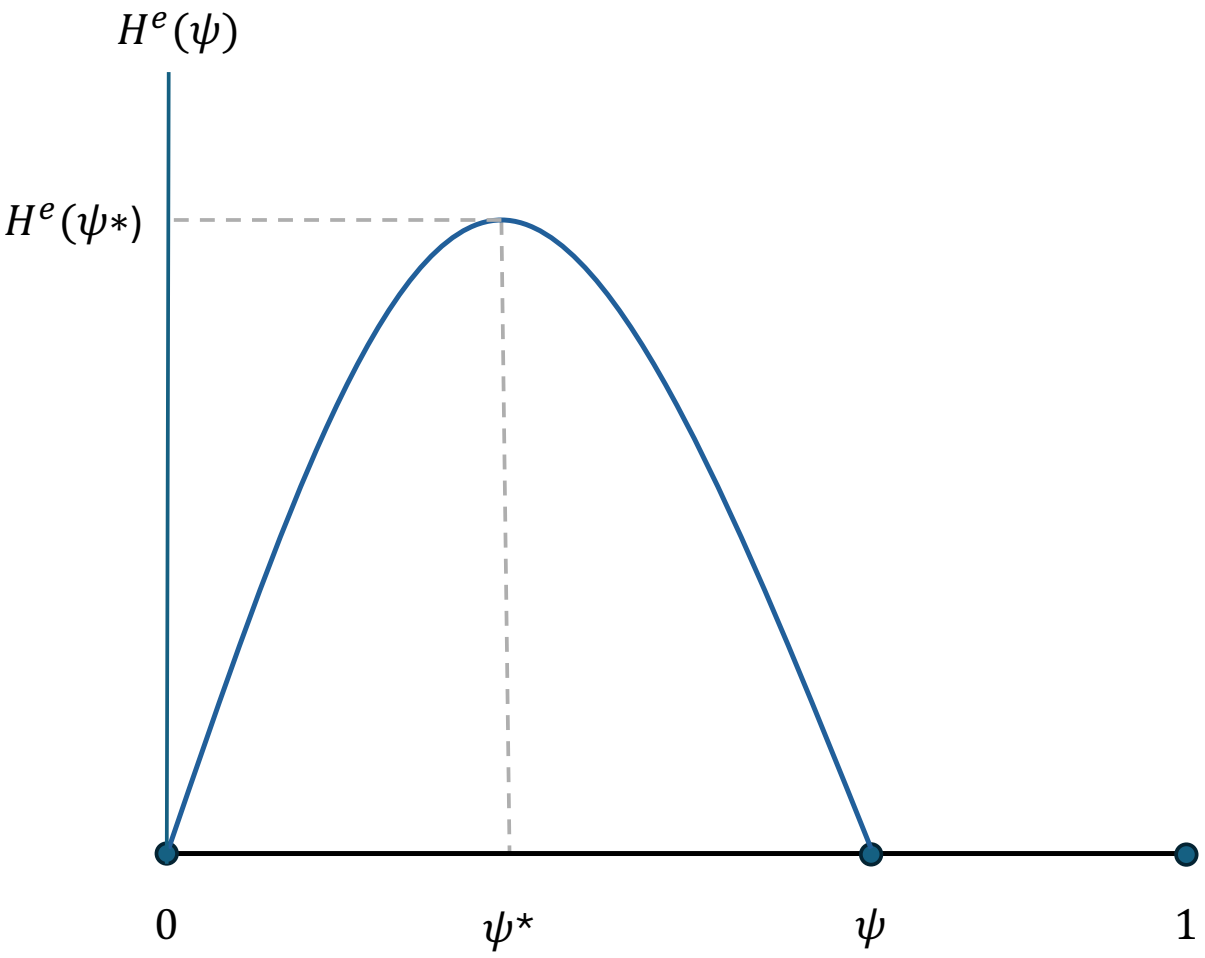
  - Earning capacity:

$$I(\psi) = w(1-\psi)\psi^{1-\theta}; \qquad \theta \in (0,1)$$

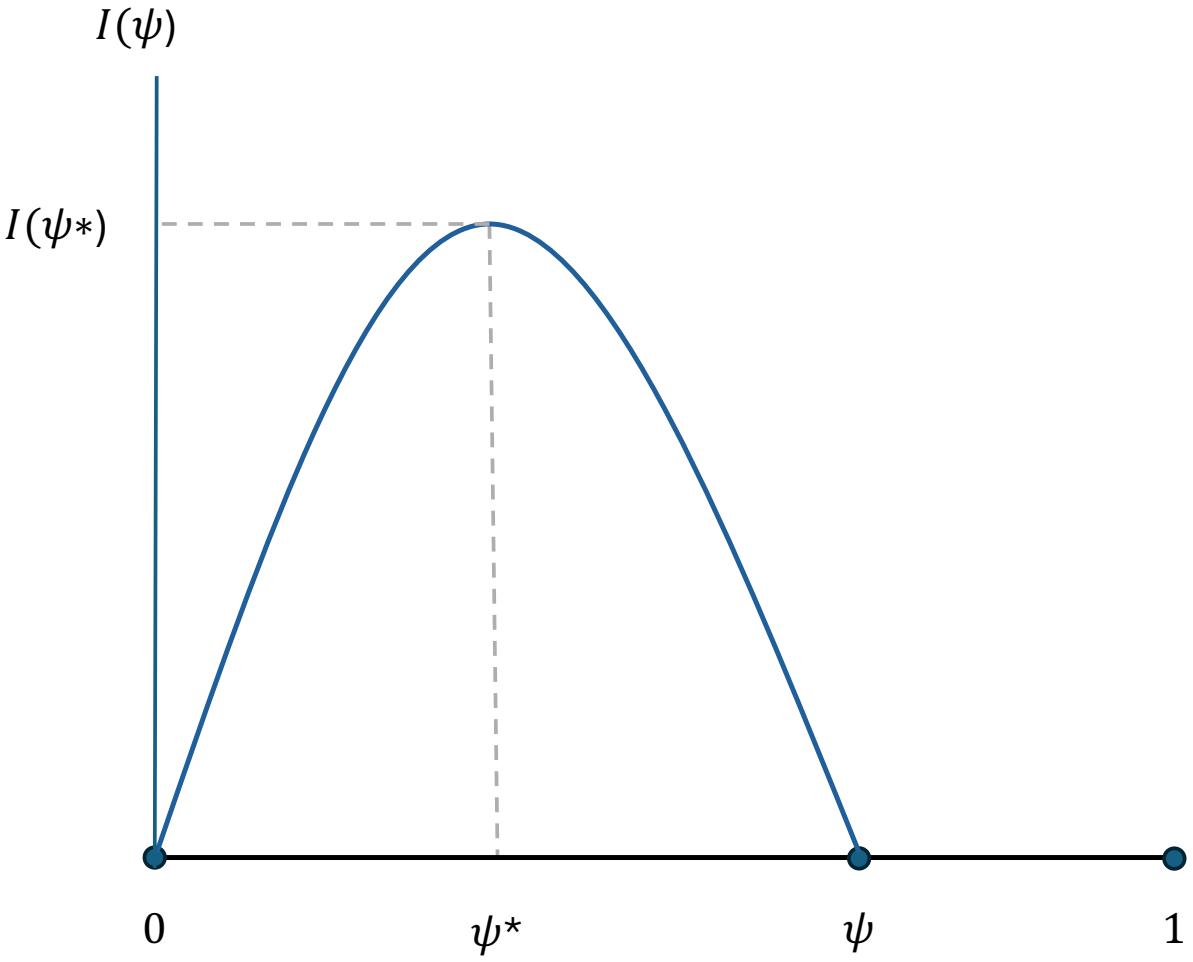
    - $\Rightarrow$ An intermediate cultural diversity, $\psi^* \in (0,1)$, maximizes $I(\psi)$

$$\psi^* = \arg\max\,[I(\psi)] = (1-\theta)/(2-\theta)$$

Diverse Mindset & Effective Human Capital



Diverse Mindset & Earning Capacity



Preferences

- Preferences:
  - $U(\psi) = (1-\alpha)\ln[H(\psi)] + \alpha \ln[(I(\psi)]; \qquad \alpha \in (0,1)$
- $\Rightarrow$ Individuals values:
  - $H(\psi)$ – Intelectual wealth
  - $I(\psi)$ – Material wealth
- $\alpha \equiv$ weight assigned to material vs. intellectual wealth
  - Reflecting (partly) the individuals's degree of:
    - Future-Oriented Mindset
    - Societal (Parental) influence

The Desirable Range of Skills

- Non-monotonic effect of ψ on the individual's earning capacity $\Rightarrow$
  - If cultural diversity ψ exceeds the income maximizing level ψ^*, $\psi > \psi^*$
    - Individuals can choose to *narrow* their acquired set of skills $\hat{\psi}$

$$\hat{\psi} = \arg\max\ U(\psi) \quad \text{s.t.} \quad \psi > \psi^*$$

  - If cultural diversity is lower than the income maximizing level, ψ^*
    - Acquiring a *wider* range of skills is prohibitively costly

The Desirable Range of Skills

$$\alpha = 1 \quad \Rightarrow \quad \hat{\psi} = \psi^*$$

$$\alpha = 0 \quad \Rightarrow \quad \hat{\psi} = \psi$$

$$\hat{\psi}'(\alpha) < 0 \qquad \text{for} \quad \alpha \in (0, 1)$$

- Individuals whose diverse mindset exceeds the income maximizing level, ψ^* &
  - Place their entire weight on their income ($\alpha = 1$)
    - Choose to narrow their acquired set of skills to ψ^*
  - Place their entire weight on knowledge formation ($\alpha = 0$)
    - Choose to acquire the entire range of their inherent skills ψ
  - Greater weight to earning capacity $\Rightarrow$ narrower set of acquired skills

Empirical Strategy

- Leverage variation in the ancestral origins of the U.S. population to estimate:
  - The impact of ancestral cultural distance on income
- Accounting for:
  - Ancestry i fixed effects
    - All observable and unobservable factors associated with ancestry i that may affect earninig

      (future-oriented mindset, gender attitudes, trust, cooperation, and individualism)
  - Gender & Age FE

Empirical Design

- Reverse causality – Not feasible
  - Individuals' wages could not affect their ancestral cultural distances
- Omitted Variables:
  - Unobserved heterogeneity among each of the ancestral origins
    - Accounted for by Ancestries FE
  - Unobserved heterogeneity across ancestral pairs
    - Matters in the unlikely event that they are correlated with the migratory distance between a pair of ancestries

Ω